REVISION NOTES

Week 1

Fri 16 – Fri 23 October 2026

34	11 h 58 min	30 / 34
LECTURES	VIDEO	SUMMARISED

COURSES THIS WEEK

MATH 110 Linear Algebra	15 lectures · 2 h 43 min
AI 200 Neural Networks: Zero to Hero	6 lectures · 4 h 48 min
MATH 120 Single-Variable Calculus	6 lectures · 1 h 38 min
SPCH 100 Storytelling	5 lectures · 1 h 17 min
MATH 130 Probability	2 lectures · 1 h 32 min
Total	**34 lectures · 11 h 58 min**

Study days Fri 16 Oct to Fri 23 Oct; Sat 17 Oct is the rest day. Hours are video time; a lecture the plan spreads over several days counts only the days that fall in this week.

WEEK 1 · FRI 16 – FRI 23 OCTOBER 2026

Contents

LECTURES, IN PLAN ORDER

1	MATH 110	Vectors		Fri 16
2	MATH 110	Linear combinations, span, and basis vectors		Fri 16
3	MATH 110	Linear transformations and matrices		Fri 16
4	MATH 110	Matrix multiplication as composition		Fri 16
5	MATH 110	Three-dimensional linear transformations		Fri 16
6	MATH 110	The determinant		Fri 16
7	AI 200	But what is a neural network?	No summary	Fri 16
8	AI 200	Gradient descent — how neural networks learn	No summary	Fri 16
9	MATH 120	The essence of calculus		Sun 18
10	MATH 120	The paradox of the derivative		Sun 18
11	MATH 120	Derivative formulas through geometry		Sun 18
12	AI 200	What is backpropagation really doing?	No summary	Sun 18
13	AI 200	Backpropagation calculus	No summary	Sun 18
14	SPCH 100	Homework for Life — finding the story in an ordinary day		Sun 18
15	MATH 130	Probability and Counting		Mon 19
16	AI 200	Micrograd — spelled-out intro to neural networks and backpropagation		Mon 19 – Wed 21
17	SPCH 100	Five ways to improve your professional voice		Mon 19
18	MATH 110	Inverse matrices, column space and null space		Tue 20
19	MATH 110	Nonsquare matrices as transformations between dimensions		Tue 20
20	MATH 110	Dot products and duality		Tue 20
21	MATH 110	Cross products		Tue 20
22	MATH 110	Cross products in the light of linear transformations		Tue 20
23	SPCH 100	Think faster, talk smarter — structure under pressure		Tue 20 – Wed 21
24	MATH 120	Visualizing the chain rule and product rule		Wed 21
25	MATH 120	What's so special about Euler's number e?		Wed 21
26	MATH 120	Implicit differentiation, what's going on here?		Wed 21
27	MATH 130	Story Proofs, Axioms of Probability		Thu 22
28	AI 200	Makemore 1 — bigram character-level language model		Thu 22 – Fri 23
29	SPCH 100	The Pixar story spine — six prompts that hold a story up		Thu 22
30	SPCH 100	How to easily be funnier in conversations		Thu 22
31	MATH 110	Cramer's rule, explained geometrically		Fri 23
32	MATH 110	Change of basis		Fri 23
33	MATH 110	Eigenvectors and eigenvalues		Fri 23
34	MATH 110	A quick trick for computing eigenvalues		Fri 23

AT THE BACK

Not yet summarised 4 lectures
What this week established
Answers to the self-checks

30 of 34 lectures have a summary. The other 4 are listed, with their video links, on “Not yet summarised”; nothing is written for them here.

MATH 110 · LINEAR ALGEBRA · LECTURE 1 OF 34

Vectors

Fri 16 Oct · 10 min video · Watch on YouTube

THE TAKEAWAY

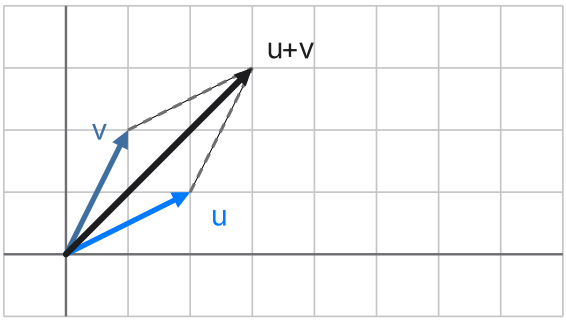
Everything in linear algebra is built from exactly two operations: adding vectors and scaling them.

THE ARGUMENT · 4 STEPS

1. **Three views of the same object**
   Physics: an arrow with length and direction, free to sit anywhere. Computer science: an ordered list of numbers. Mathematics: anything at all that you can add and scale. The whole subject lives in translating between the arrow and the list.

2. **Coordinates are instructions, not identity**
   Root the arrow at the origin and its coordinates say: walk this far along x, then this far up y. The numbers describe the arrow relative to a chosen frame — they are not the arrow itself.

3. **Addition is composing two movements**
   Slide the second arrow so its tail meets the first arrow's tip; the sum runs from the original tail to the final tip. That is why addition works out componentwise — you walked x_1 then x_2 in the same direction.



4. **Scaling stretches, squashes, or flips**
   Multiplying by 2 doubles the length, by ½ halves it, by −1 reverses the direction. This is exactly why numbers in this context are called scalars: their job is to scale.

WORKED EXAMPLE

$(1, 2) + (3, -1)$. Tip-to-tail: walk 1 right and 2 up, then 3 right and 1 down. You end at $(4, 1)$ — the same answer as adding each coordinate separately. The geometry and the arithmetic are the same fact.

WATCH FOR

Treating the coordinate pair as the vector. Change the basis and every coordinate changes while the arrow does not move at all.

KEY CONCEPTS

A vector is a displacement

CHECK YOURSELF · ANSWERS AT THE BACK

1.1 Vector addition works componentwise because:

A It is defined that way by convention
B Walking one displacement then the other adds each direction separately
C Vectors must start at the origin
D Only the lengths matter

1.2 The first component of $(1, 2) + (3, -1)$ is:

Your answer: _____

MATH 110 · LINEAR ALGEBRA · LECTURE 2 OF 34

Linear combinations, span, and basis vectors

Fri 16 Oct · 10 min video · Watch on YouTube

THE TAKEAWAY

Coordinates are scalars applied to a chosen basis, and the span is everything those scalars can reach.

THE ARGUMENT · 5 STEPS

1 **The basis is hiding inside the coordinates**

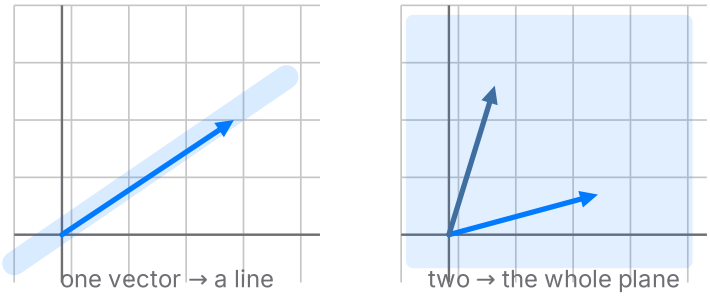
The pair $(3, 2)$ silently means $3\hat{\imath} + 2\hat{\jmath}$. Choose different basis vectors and the same pair of numbers points somewhere else entirely — coordinates only mean something relative to a basis.

2 **Linear combination**

Fix two vectors, let their scalars roam over every value, and the set of results is the set of linear combinations. The word 'linear' is earned: if you hold one scalar fixed and vary the other, the tip traces a straight line.

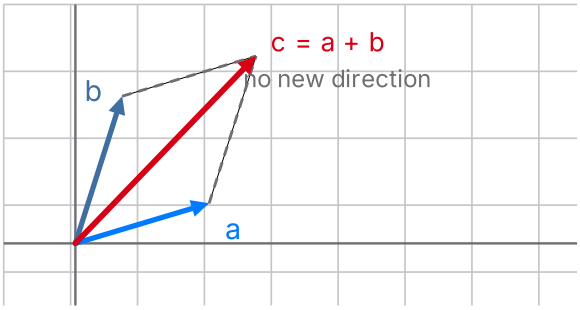
3 Span is the reachable set

Two non-collinear vectors in the plane reach every point in it. Two collinear ones reach only a single line, because the second offers no direction the first did not already have.



4 Dependence is redundancy

A vector is dependent on the others when it already lies in their span — removing it shrinks nothing. Independent means every vector contributes a direction genuinely its own.



5 Basis, defined properly

A basis is an independent set that spans the space: enough vectors to reach everywhere, few enough that none is wasted.

WORKED EXAMPLE

Do $(1,2)$ and $(2,4)$ span the plane? Ask whether one is a multiple of the other: $(2,4) = 2(1,2)$, so they are collinear and span only a line. Contrast $(1,2)$ and $(0,1)$, which are not multiples and so reach every point.

WATCH FOR

Assuming independent means perpendicular. Two vectors one degree apart are independent — awkward to compute with, but independent.

KEY CONCEPTS

- Span is everywhere you can reach
- Independence means no wasted direction
- A basis is the ruler you measure with

CHECK YOURSELF · ANSWERS AT THE BACK

2.1 The span of two collinear vectors in $\mathbb{R}^2$ is:

A The whole plane
B A line through the origin
C Two lines
D The origin only

2.2 A basis of $\mathbb{R}^3$ contains how many vectors?

Your answer: _____

MATH 110 · LINEAR ALGEBRA · LECTURE 3 OF 34

Linear transformations and matrices

Fri 16 Oct · 11 min video · Watch on YouTube

THE TAKEAWAY

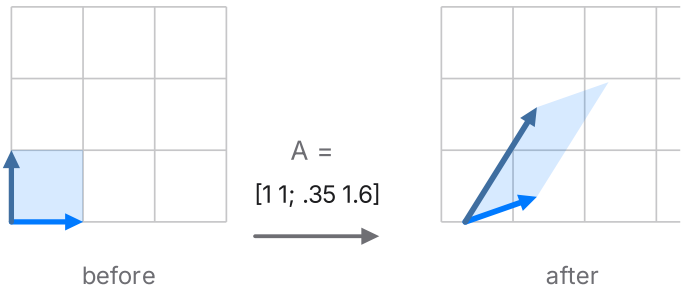
A linear transformation is fully determined by where the basis vectors land — and that is all a matrix records.

THE ARGUMENT · 4 STEPS

1. **What makes a transformation linear**
   Two visual conditions: every line stays a line (never curved), and the origin stays put. Equivalently, grid lines stay parallel and evenly spaced.

2. **The consequence that does all the work**
   Because grid lines stay parallel and evenly spaced, a vector that was $x\hat{\imath} + y\hat{\jmath}$ before is $x(\text{where } \hat{\imath} \text{ landed}) + y(\text{where } \hat{\jmath} \text{ landed})$ after. Track two vectors and you know the fate of every vector.

3. **The matrix is just those landing spots**
   Write where $\hat{\imath}$ landed in the first column and where $\hat{\jmath}$ landed in the second. That is the matrix. Nothing else is going on.



4 **So multiplication is a linear combination of columns**

$\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = x\begin{pmatrix} a \\ c \end{pmatrix} + y\begin{pmatrix} b \\ d \end{pmatrix}$ — scale each column by the matching coordinate and add.

WORKED EXAMPLE

Build a 90° counter-clockwise rotation from scratch. $\hat{\imath} = (1, 0)$ rotates to $(0, 1)$; $\hat{\jmath} = (0, 1)$ rotates to $(-1, 0)$. Put those in as columns: $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. You never memorised it — you read it off the geometry.

WATCH FOR

Reading a matrix as a grid of numbers to manipulate. Read the columns as destinations and most matrix facts become obvious instead of memorised.

KEY CONCEPTS

A matrix moves space

CHECK YOURSELF · ANSWERS AT THE BACK

3.1 A linear transformation is pinned down entirely by:

A Its determinant
B Where the basis vectors land
C Its eigenvalues
D Its trace

3.2 For the 90° counter-clockwise rotation, the second component of the first column is:

Your answer: _____

MATH 110 · LINEAR ALGEBRA · LECTURE 4 OF 34

Matrix multiplication as composition

Fri 16 Oct · 10 min video · Watch on YouTube

THE TAKEAWAY

Matrix multiplication is doing one transformation after another, which is why order matters and why the rows-times-columns rule looks the way it does.

THE ARGUMENT · 5 STEPS

1 **Composition produces a new transformation**

Apply a shear, then a rotation. The net effect is itself linear — lines stayed lines, the origin stayed fixed — so it has a matrix of its own. That matrix is the product.



2 **Read right to left**

AB means apply B first, then A, inherited from function notation $f(g(x))$. The right-hand matrix touches the vector first.

3 **Computing it is just tracking the basis again**

Send $\hat{\imath}$ through B, then through A: the result is the first column of AB. Repeat with $\hat{\jmath}$ for the second column. The row-times-column recipe is this bookkeeping written out.

4 **Order genuinely changes the outcome**

Shear-then-rotate and rotate-then-shear leave the basis in different places, so $AB \neq BA$. No algebra needed — watch where $\hat{\imath}$ ends up.

5 **Associativity is free**

$(AB)C = A(BC)$ needs no computation: both sides mean apply C, then B, then A. Same sequence of actions, therefore the same transformation.

WORKED EXAMPLE

Shear $S = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ then rotate $R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. Track $\hat{\imath}$: S leaves $(1, 0)$, then R sends it to $(0, 1)$. Track $\hat{\jmath}$: S sends $(0, 1)$ to $(1, 1)$, then R sends that to $(-1, 1)$. So $RS = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}$.

WATCH FOR

Expecting $AB = BA$. Rotating a sheared square is not shearing a rotated one, and the pictures differ visibly.

KEY CONCEPTS

Multiplication is composition

CHECK YOURSELF · ANSWERS AT THE BACK

4.1 In AB, which transformation acts on the vector first?

A A
B B
C Both simultaneously
D Whichever has larger entries

4.2 $(AB)C = A(BC)$ holds because:

A Determinants multiply
B Both describe the same sequence of actions
C Matrices commute
D It only holds for square matrices

MATH 110 · LINEAR ALGEBRA · LECTURE 5 OF 34

Three-dimensional linear transformations

Fri 16 Oct · 5 min video · Watch on YouTube

THE TAKEAWAY

Nothing changes in three dimensions: the transformation is still completely described by where the basis vectors land.

THE ARGUMENT · 5 STEPS

1. **The same two conditions**
   Grid lines stay parallel and evenly spaced, and the origin stays put. The grid is now a lattice filling space rather than a sheet, but the definition of linear does not gain a clause.

2. **Three basis vectors, not two**
   $\hat{\imath}$ along x, $\hat{\jmath}$ along y, and a new one, $\hat{k}$, along z. Follow only those three — the full moving grid is noise, and the three landing spots carry all the information.

3. **Nine numbers is just three columns**
   Write where each basis vector lands as a column. Three vectors, three coordinates each: a 3×3 matrix, and it describes the whole transformation.

4. **Scale-and-add survives the transformation**
   The coordinates (x, y, z) are instructions for scaling the basis vectors and adding them. That recipe holds before and after, which is exactly why you scale each column by the matching coordinate and sum.

5 Composition is still composition

Two 3×3 matrices multiplied means: apply the right transformation, then the left. Graphics and robotics lean on this — a rotation that is hard to picture is built from separate rotations that are not.

WORKED EXAMPLE

Rotate space 90° about the y-axis. $\hat{\imath} = (1, 0, 0)$ lands on $(0, 0, -1)$; $\hat{\jmath}$ does not move; $\hat{k}$ lands on $(1, 0, 0)$. Those three triples are the columns. You read the matrix off the geometry instead of recalling a formula.

WATCH FOR

Assuming three dimensions needs new machinery. It needs one more basis vector and one more coordinate; every rule from two dimensions carries over untouched.

KEY CONCEPTS

A matrix moves space

CHECK YOURSELF · ANSWERS AT THE BACK

5.1 A 3×3 matrix has nine entries because:

A Three dimensions squared is nine
B Three basis vectors each need three coordinates to say where they landed
C A matrix must be square to be invertible
D Each row is one equation

5.2 Rotating 90° about the y-axis sends $\hat{\imath}$ to $(0, 0, -1)$. The third component of the first column is:

Your answer: _____

MATH 110 · LINEAR ALGEBRA · LECTURE 6 OF 34

The determinant

Fri 16 Oct · 10 min video · Watch on YouTube

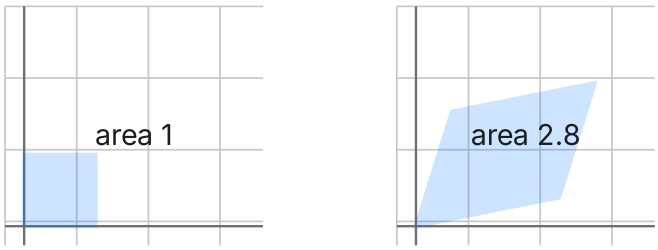
THE TAKEAWAY

The determinant is the factor by which a transformation scales area, and its sign says whether space was flipped.

THE ARGUMENT · 6 STEPS

1 **Measure the transformation by what it does to one square**

The unit square spanned by $\hat{\imath}$ and $\hat{\jmath}$ has area 1. After the transformation it is a parallelogram, and its area is the determinant. Because grid lines stay parallel and evenly spaced, every other region scales by that same factor.



unit square det A = 2.8

2 **Zero is the interesting case**

A determinant of zero means the square was flattened onto a line (or a point): the columns became dependent, dimension was lost, and information was destroyed. That is precisely when the matrix has no inverse.

3 **Negative means orientation flipped**

If the transformation turns space over — $\hat{\jmath}$ ending up clockwise from $\hat{\imath}$ rather than counter-clockwise — the determinant is negative. Its magnitude is still the area factor.

4 **Where $ad - bc$ comes from**

For $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ the parallelogram's area works out to $ad - bc$. With $b = c = 0$ it is a plain a-by-d rectangle; the bc term corrects for how much the shape has been sheared.

5 **In three dimensions it is volume**

The unit cube becomes a parallelepiped and the determinant is its volume, with sign given by whether the basis stayed right-handed.

6 **Products multiply**

$\det(M_1 M_2) = \det(M_1)\det(M_2)$, and no computation is required to believe it: apply one area factor and then the other.

WORKED EXAMPLE

$\det \begin{pmatrix} 3 & 0 \\ 0 & 2 \end{pmatrix} = 6$ — a 3-by-2 rectangle, area 6. Now $\det \begin{pmatrix} 1 & 1 \\ 2 & 2 \end{pmatrix} = 1 \cdot 2 - 1 \cdot 2 = 0$: the columns are collinear, the square collapses onto a line, and nothing can be inverted.

WATCH FOR

Thinking the determinant measures how big a matrix is. It measures how much the map multiplies volume — a matrix of huge entries can have determinant zero.

KEY CONCEPTS

The determinant is a volume factor

CHECK YOURSELF · ANSWERS AT THE BACK

6.1 $\det A = 0$ tells you the transformation:

A Preserves area
B Collapses space into a lower dimension
C Is a rotation
D Has no eigenvalues

6.2 $\det \begin{pmatrix} 3 & 0 \\ 0 & 2 \end{pmatrix}$ is:

Your answer: _____

AI 200 · NEURAL NETWORKS: ZERO TO HERO · LECTURE 7 OF 34

But what is a neural network?

Fri 16 Oct · 19 min video · Watch on YouTube

NOT YET SUMMARISED

This lecture has no summary yet, so nothing is printed for it. It is listed with its video link on “Not yet summarised” at the back.

AI 200 · NEURAL NETWORKS: ZERO TO HERO · LECTURE 8 OF 34

Gradient descent — how neural networks learn

Fri 16 Oct · 21 min video · Watch on YouTube

NOT YET SUMMARISED

This lecture has no summary yet, so nothing is printed for it. It is listed with its video link on “Not yet summarised” at the back.

MATH 120 · SINGLE-VARIABLE CALCULUS · LECTURE 9 OF 34

The essence of calculus

Sun 18 Oct · 17 min video · Watch on YouTube

THE TAKEAWAY

Hard problems become sums of many small quantities, those sums become areas under graphs, and derivatives are how you find those areas.

THE ARGUMENT · 6 STEPS

1. **Slice the circle in a way that respects its symmetry**
   Concentric rings, not squares. A ring at radius r unrolls into something close to a rectangle: width $2\pi r$, which is essentially the definition of π, and thickness dr, which is however finely you chose to slice.

2. **The approximation gets better exactly as dr shrinks**
   The unrolled ring is not a rectangle — its top and bottom edges differ. But they differ less the thinner the ring, so $2\pi r\,dr$ is wrong by an amount that vanishes. Holding that fact in mind is the whole method.

3. **Stand the rectangles side by side and a graph appears**
   Each has width dr, so they pack snugly along a r-axis, and each has height $2\pi r$. Their tops trace the line $y = 2\pi r$. The sum of many small numbers has become the area under a graph.

4. **That area is a triangle, and out falls πr^2**
   Base 3, height $2\pi \cdot 3$, so $\frac{1}{2} \cdot 3 \cdot 6\pi = 9\pi$. For a general radius R it is πR^2 — the formula, derived rather than remembered.

5. **Now ask the same question of a curve that is not a triangle**
   What is the area under $y = x^2$ from 0 to x? Call it $A(x)$ — the INTEGRAL of x^2. It is a mystery function, and attacking it head-on gets you nowhere.

6. **So nudge x and watch what the area does**
   Push x by dx and the new sliver is nearly a rectangle: height x^2, width dx. So $dA \approx x^2\,dx$, which rearranges to $\frac{dA}{dx} \approx x^2$. That ratio is the DERIVATIVE, and the relation between it and the area is the fundamental theorem.

WORKED EXAMPLE

The method transfers. Distance from velocity: at each moment multiply the velocity by a tiny dt to get a tiny distance, then add them. Any problem that becomes a sum of thin rectangles side by side is a problem about the area under a graph.

WATCH FOR

Treating $2\pi r\,dr$ as the ring's exact area. It is an approximation, and the argument works because the error shrinks faster than the number of rings grows — not because the rectangle was ever right.

CHECK YOURSELF · ANSWERS AT THE BACK

9.1 Summing the ring areas $2\pi r\,dr$ turns into an area under a graph because:

A Circles and triangles have the same area

B Each term is a rectangle of width dr and height $2\pi r$, and they pack side by side under $y = 2\pi r$

C π is irrational

D The rings are exactly rectangles

9.2 For $A(x)$, the area under $y = x^2$ from 0 to x, the ratio dA/dx at $x = 3$ is:

Your answer: _____

MATH 120 · SINGLE-VARIABLE CALCULUS · LECTURE 10 OF 34

The paradox of the derivative

Sun 18 Oct · 17 min video · Watch on YouTube

THE TAKEAWAY

"Instantaneous rate of change" is an oxymoron, and the derivative is the clever construction that gets the meaning without the contradiction.

THE ARGUMENT · 6 STEPS

1. **Velocity at a single instant is not a thing**
   Show someone a photograph of a car and ask its speed — there is no answer. Speed needs two moments to compare: a change in distance over a change in time. Yet a velocity graph assigns a number to each single t. That tension is the whole chapter.

2. **What a real speedometer does is sidestep it**
   It measures how far the car goes between 3 and 3.01 seconds and divides by 0.01. Never a single instant — always a small interval. Call them ds and dt, and velocity is ds/dt.

3. **On the graph that ratio is a slope**
   dt is a step right, ds is the resulting rise, so ds/dt is rise over run between two nearby points on the distance curve. Steep graph, fast car; shallow graph, slow car.

4. **The derivative is what the ratio APPROACHES**
   Not the ratio at some particular dt, and not $dt = 0$ — that is division by zero. As the two points slide together, the secant line approaches the TANGENT line, and the derivative is that tangent's slope.

5. **Letting $dt \to 0$ makes the algebra simpler, not harder**
   For $s = t^3$ at $t = 2$: expanding $(2 + dt)^3 - 2^3$ gives a mess, but the 2^3 cancel, every surviving term has a dt, and after dividing only $3 \cdot 2^2$ has no dt left. Everything else vanishes. So $\frac{d}{dt}t^3 = 3t^2$.

6 **The paradox, sharpened**

For $s = t^3$ the derivative at $t = 0$ is 0. So is the car moving? Between 0 and 0.1 seconds it travels 0.001 m — it moves. The derivative being 0 does not mean static; it means the best CONSTANT approximation to its speed near 0 is 0.

WORKED EXAMPLE

Notation warning: the d in ds/dt announces an intention. It says you are going to ask what the ratio approaches as the nudge shrinks. It is not a fraction, even though it behaves like one often enough to be misleading.

WATCH FOR

Reading the derivative as a rate of change at an instant. Change needs two moments. Read it instead as the best constant approximation for the rate of change near a point — which is what the tangent line is.

CHECK YOURSELF · ANSWERS AT THE BACK

10.1 The derivative is not simply ds/dt for a small dt because:

A dt would be negative

B It is what that ratio APPROACHES as dt shrinks — a limit, not a value at any particular dt

C ds is always zero

D Small numbers are hard to compute

10.2 For $s(t) = t^3$, the derivative at $t = 2$ is:

Your answer: _____

MATH 120 · SINGLE-VARIABLE CALCULUS · LECTURE 11 OF 34

Derivative formulas through geometry

Sun 18 Oct · 18 min video · Watch on YouTube

THE TAKEAWAY

Derivative formulas are not rules to memorise: draw what the function literally means, nudge the input, and read the answer off the picture.

THE ARGUMENT · 6 STEPS

1 **x^2 is a square, so nudge its side**

Grow the side by dx and three new pieces appear: two thin rectangles of area $x\,dx$ each, and one tiny corner square of area dx^2. So $df \approx 2x\,dx$, and $df/dx = 2x$.

2 **The corner square is why the rule is clean**

At $dx = 0.01$ the two rectangles contribute 0.06 and the corner contributes 0.0001. Anything with dx raised above the first power is negligible — that is the rule of thumb the whole subject leans on.

3 **x^3 is a cube, and the same thing happens**

Three thin square slabs of volume $x^2\,dx$ each, plus edge slivers and a corner cube that all carry dx^2 or dx^3. So $\frac{d}{dx}x^3 = 3x^2$ — which the graph alone could never have told you.

4 **The power rule is that argument in general**

Expanding $(x + dx)^n$ gives x^n, then n terms that pick a single dx from one of the n parentheses — $n\,x^{n-1}dx$ — then everything else, all carrying dx^2 or worse. So $\frac{d}{dx}x^n = nx^{n-1}$.

5 **$1/x$ is a puddle of fixed area**

Width x, height $1/x$, area always 1. Widen it by dx and the height must drop by exactly enough to cancel the area gained. That drop is negative, which is where the minus sign in $-1/x^2$ comes from.

6 **$\sin\theta$ is a height on the unit circle**

Step $d\theta$ along the circumference. Zoomed in, the arc is a straight line and the little triangle is SIMILAR to the big one with angle θ and hypotenuse 1. So $d(\sin\theta)/d\theta$ is adjacent over hypotenuse — the definition of $\cos\theta$.

WORKED EXAMPLE

The graph tells you the shape, the meaning tells you the formula. The slope of $\sin$ is clearly positive at 0, zero at the peak, negative after — and that is true of infinitely many functions. Only the unit-circle triangle pins it to exactly $\cos\theta$.

WATCH FOR

Keeping the dx^2 term because it looks like it should count. It is divided by dx along with everything else, so one factor of dx survives — and then vanishes as $dx \to 0$. Second-order terms never reach the answer.

CHECK YOURSELF · ANSWERS AT THE BACK

11.1 The derivative of $\sin\theta$ is exactly $\cos\theta$ rather than merely something with the same peaks because:

A Both graphs are waves
B The nudge triangle on the unit circle is similar to the angle-θ triangle, so the ratio IS adjacent over hypotenuse
C Sine and cosine are inverses
D π appears in both

11.2 By the power rule, the derivative of x^5 at $x = 2$ is:

Your answer: _____

AI 200 · NEURAL NETWORKS: ZERO TO HERO · LECTURE 12 OF 34

What is backpropagation really doing?

Sun 18 Oct · 13 min video · Watch on YouTube

NOT YET SUMMARISED

This lecture has no summary yet, so nothing is printed for it. It is listed with its video link on “Not yet summarised” at the back.

AI 200 · NEURAL NETWORKS: ZERO TO HERO · LECTURE 13 OF 34

Backpropagation calculus

Sun 18 Oct · 10 min video · Watch on YouTube

NOT YET SUMMARISED

This lecture has no summary yet, so nothing is printed for it. It is listed with its video link on “Not yet summarised” at the back.

SPCH 100 · STORYTELLING · LECTURE 14 OF 34

Homework for Life — finding the story in an ordinary day

Sun 18 Oct · 18 min video · Watch on YouTube

THE TAKEAWAY

Homework for Life: at the end of each day, ask what the most story-like moment was and write a line or two. Done daily it builds a storyteller's lens, fills a bank of small stories, and — the part Dicks did not expect — slows time down.

THE ARGUMENT · 7 STEPS

1. **The lunch box**
   When the doctor says to take food away from a food-throwing toddler, his wife warns it will be hard for him because he grew up hungry — a secret he never told her — and admits she quietly unpacks the oversized lunches he packs every morning.

2. **Dander in the wind**
   Moments like that are everywhere. We miss most of them, and when we do see one we have no way of catching it.

3. **Small stories connect**
   His dramatic stories — dying, jail, homelessness — are only okay on stage. The small ones are what audiences love, because everyone has a secret and everyone knows hunger.

4. **The assignment**
   Every night: if I had to tell a five-minute story from something that happened today, however benign, what was the most story-like moment? Write a sentence or two. Not the story.

5. **The lens sharpens**
   Walking an old dog at 2 a.m., in boxers, in the rain, becomes a moment he will own forever instead of an annoyance. Forgotten childhood memories start surfacing too.

6. **Time slows down**
   Days stop blurring together. He can touch any moment from years of entries and be back inside it.

7. **Commitment and faith**
   Five minutes every day, and patience: the early entries are weak, and the lens can take months to sharpen.

WORKED EXAMPLE

Same time, same place, every night, one line per moment however ordinary — 'walked Kaylee at 2 a.m., underwear, birds, rain, beauty'. A few weeks later, reread the list and mark the entries that still pull you straight back into the moment.

WATCH FOR

Waiting for something dramatic before writing anything down. The practice depends on logging the ordinary; skipping the 'boring' days is exactly what keeps the lens from sharpening.

CHECK YOURSELF · ANSWERS AT THE BACK

14.1 What question does Dicks ask himself each night?

A What was the most dramatic thing that happened to me today, worth a full story?
B What was the most story-like moment of today, if I had to tell a story from it?
C What did I learn today that I could teach in a five-minute talk?
D Which conversation today would make the funniest story at dinner?

14.2 Why doesn't he write the whole story down?

A It would be too much work, and he would stop doing it
B Writing it out would fix the story before it was ready
C He keeps the full stories in a separate notebook instead
D Stories should be told aloud, never written

14.3 Which effect of the practice did he not expect?

A Winning more StorySLAMs with smaller, quieter stories
B Sleeping better after writing each night
C Time slowing down, so days stop disappearing
D Finding his dramatic stories went down better

MATH 130 · PROBABILITY · LECTURE 15 OF 34

Probability and Counting

Mon 19 Oct · 46 min video · Watch on YouTube

THE TAKEAWAY

The naive definition — favourable outcomes over possible outcomes — is correct only when the outcomes are equally likely and finite in number, and the rest of this lecture is learning to count the two numbers.

THE ARGUMENT · 12 STEPS

1. **Statistics is the logic of uncertainty**
   Mathematics is the logic of certainty. Everyone has uncertainty, and probability is how you quantify it and update it. That is the whole subject in one sentence.

2. **It started with gambling, and the intuitions were wrong**
   Fermat and Pascal worked it out in letters to each other in the 1650s, analysing games of chance. Newton was consulted too, and although his arithmetic was right, his INTUITION about a dice problem was wrong. Expect to be surprised often.

3 **Sample space and event**

A sample space S is the set of all possible outcomes of an experiment — read 'experiment' as broadly as you like. An EVENT is a subset of S. Treating events as sets was the breakthrough that made probability mathematics rather than a collection of heuristics.

4 **The naive definition**

$P(A) = \dfrac{\text{number of outcomes favourable to } A}{\text{number of possible outcomes}}$. Two coins give $\{HH, HT, TH, TT\}$, so $P(\text{both tails}) = \frac{1}{4}$.

5 **Its two assumptions, both strong**

All outcomes equally likely, and finitely many of them. Symmetry sometimes justifies the first — a cube has six identical faces — but a loaded die has six outcomes too.

6 **Life on Neptune**

Either there is or there is not, so the naive definition says $\frac{1}{2}$. Then intelligent life on Neptune is also $\frac{1}{2}$ — which cannot be, since it must be strictly less likely. Listing two cases is not the same as showing they are equally likely, and this argument appears in the news regularly.

7 **The multiplication rule**

If experiment 1 has n_1 outcomes and, whatever happened there, experiment 2 has n_2, and so on, the combined experiment has $n_1 n_2 \cdots n_r$ outcomes. Picture the tree: two cone types branching into three flavours is six leaves, and $2 \times 3 = 3 \times 2$, so the order you branch in does not matter.

8 **Which is why listing is hopeless**

Ten binary choices already give $2^{10} = 1024$ leaves. Counting has to replace enumerating almost immediately.

9 **The binomial coefficient, derived not memorised**

$\binom{n}{k} = \frac{n!}{(n-k)!\,k!}$, and it is 0 when $k > n$. Pick in order — $n(n-1)\cdots(n-k+1)$ ways — then divide by $k!$, because you counted every subset once per ordering.

10 **A full house, by the tree**

Choose the rank you hold three of (13), choose which three of its four suits $\binom{4}{3}$, choose the other rank (12, not 13), choose two of its suits $\binom{4}{2}$. Over $\binom{52}{5}$ hands, that is about 0.00144.

11 **The sampling table**

Choosing k from n: with replacement and order mattering, n^k; without replacement and order mattering, $n(n-1)\cdots(n-k+1)$; without replacement and order not mattering, $\binom{n}{k}$. All three follow straight from the multiplication rule.

12 **The fourth cell is much harder than the other three**

With replacement and order NOT mattering, the count is $\binom{n+k-1}{k}$ — and nothing about the multiplication rule makes that obvious. It is worth checking on tiny cases before seeing the proof.

WORKED EXAMPLE

Write down a concrete instance before counting anything. 'Three sevens and two tens' is what makes the full-house count structured: it tells you there are two ranks to choose, in a definite role, and that the second choice has 12 options rather than 13.

WATCH FOR

Applying the naive definition because the outcomes are easy to list. The question is never how many cases there are but whether they are equally likely — and a sentence saying why should appear in the solution, not just the fraction.

CHECK YOURSELF · ANSWERS AT THE BACK

15.1 Why is 'either there is life on Neptune or there is not, so $P = \frac{1}{2}$' wrong?

A Because the sample space is infinite
B Because listing two outcomes does not make them equally likely, and the naive definition requires that
C Because probability cannot apply to astronomy
D Because the events are not disjoint

15.2 The probability of a full house in a five-card hand, to five decimal places, is:

Your answer: _____

15.3 Choosing 3 items from 10 with replacement, order not mattering, the number of possibilities is:

Your answer: _____

AI 200 · NEURAL NETWORKS: ZERO TO HERO · LECTURE 16 OF 34

Micrograd — spelled-out intro to neural networks and backpropagation

Mon 19 Oct – Wed 21 Oct · day 1–3 of 3 · 146 min video · Watch on YouTube

THE TAKEAWAY

Backpropagation is the chain rule applied node by node over an expression graph: each operation knows only its local derivative, gradients flow from the output back through a topological order, and they ADD wherever a value is used more than once — about a hundred lines of Python, and the same numbers PyTorch gives.

THE ARGUMENT · 10 STEPS

1. **A derivative is a nudge**
   Bump an input by a tiny h and see how much the output moves: $(f(x+h) - f(x))/h$. For $f(x) = 3x^2 - 4x + 5$ the slope is about 0 at $x = 2/3$. For $d = ab + c$, nudging c by h moves d by exactly h — the slope with respect to c is 1.

2. **Value: a number that remembers where it came from**
   A Value wraps a scalar and records its children and the operation that produced it, so a calculation builds an expression graph as it runs. Each node also carries grad — the derivative of the final output with respect to it — starting at 0.

3. **Backprop by hand**
   Set the output's grad to 1 and walk backwards. A + node passes its gradient to both inputs unchanged; a * node gives each input the other input's value times the gradient. That is the chain rule, applied locally. Nudging every leaf a little in the direction of its gradient raises L from -8 to about -7.29, as predicted.

4. **A neuron, and tanh's local derivative**
   A neuron computes $\tanh(w \cdot x + b)$. Its local derivative is $1 - o^2$, so at $o = 0.7071$ it is 0.5. Each operation gets a _backward closure that applies exactly its own local rule.

5. **backward(): topological order, then reverse**
   Build a topological ordering of the graph, set the root's grad to 1, and call each node's _backward in reverse order — so a node passes gradient to its inputs only once every node that uses it has passed gradient into it.

6. **Gradients accumulate**
   If a value is used twice — b = a + a — both paths contribute, and a.grad must be 2, not 1. So every _backward uses +=, never =. The notebook flags the one place the video got this wrong (the exp backward) and fixes it.

7. **How finely to break operations is a choice**
   Rewriting tanh as $(e^{2n} - 1)/(e^{2n} + 1)$ with exp, power, subtraction and division gives the same forward value and the same gradients. Supporting constants takes a few small methods (__radd__, __rmul__, __neg__, __sub__, __truediv__ via a power of -1).

8. **PyTorch says the same**
   The same neuron in PyTorch, with requires_grad on scalar tensors, gives o = 0.7071 and the same gradients for x1, w1, x2, w2. PyTorch works on tensors; micrograd on single numbers — which is why micrograd chops each neuron into individual adds and multiplies.

9. **A tiny neural net library, and training it**
   Neuron, Layer and MLP, each with parameters(). MLP(3, [4, 4, 1]) is trained on four examples: forward pass, loss = sum of squared errors, reset every grad to 0, loss.backward(), then step each parameter by -0.1 times its gradient. The loss falls and the predictions approach the targets.

10 **The micrograd repository**

About 100 lines of engine and 50 of nn, with ReLU neurons. Its demo trains MLP(2, [16, 16, 1]) — 337 parameters — on the two-moons data with a max-margin loss plus L2 regularization and SGD with a decaying learning rate, reaching 100% training accuracy; its tests check forward values and gradients against PyTorch.

WORKED EXAMPLE

To trust a gradient, check it two ways. Compute it with backward(), then nudge that one input by a small h and measure $(L(x+h) - L(x))/h$ — the notebook's lol() does exactly this and gets -3.99999 for dL/db where backprop says -4. Then, for a whole model, compare against PyTorch on the same inputs, as the repository's tests do.

WATCH FOR

Forgetting that gradients accumulate. Because _backward adds into grad, a training loop that does not reset every parameter's grad to 0 before loss.backward() steps on the sum of all past gradients — and writing = instead of += in a _backward silently drops every path but the last.

CHECK YOURSELF · ANSWERS AT THE BACK

16.1 The lecture starts from $f(x) = 3x^2 - 4x + 5$ and estimates slopes by nudging x by a tiny h. What is the slope of f at $x = 3$?

Your answer: _____

16.2 During backprop, a + node receives gradient g from above. What does it pass to each of its two inputs?

A g to each input, unchanged, since a sum's local derivative is 1
B g/2 to each input, so that the total gradient is preserved
C g times the other input's value, as for a multiplication
D Nothing at all: a sum has no parameters, so it stops the gradient

16.3 With a = 2, b = -3, c = 10, f = -2 and L = (a*b + c)*f, what is dL/db?

Your answer: _____

SPCH 100 · STORYTELLING · LECTURE 17 OF 34

Five ways to improve your professional voice

Mon 19 Oct · 11 min video · Watch on YouTube

THE TAKEAWAY

Half of a first impression is vocal, and most professionals switch into a flat 'professional' voice. Giang's five fixes: respect what the voice does, build a taste for voices, look in the vocal mirror, read children's books aloud, and record yourself regularly.

THE ARGUMENT · 6 STEPS

1. **Vocal image**
   Visual image is only half the first-impression puzzle. The other half forms the moment you open your mouth: how confident, friendly and trustworthy you seem.

2. **The professional voice**
   Ask a child to act like an adult and they go serious and monotone. We do the same at work, and become less engaging and less influential.

3. **Same words, opposite meanings**
   'Wow, that's a great career path' — amazed, sarcastic, defeated, and a disappointed mother. The meaning lives beneath the words.

4. **Build your vocal taste**
   Watch the most-viewed TED talks, notice whose voice draws you in and whose annoys you, and ask why — the way you develop a sense of style.

5. **The vocal mirror**
   Record yourself, watch it back, and list what you like and what the voice says about you that you do not intend.

6. **Children's books and monthly videos**
   Children's books are full of emotive words that force inflection. One recorded video a month gives twelve markers a year, so invisible progress becomes visible.

WORKED EXAMPLE

Record sixty seconds of how you normally explain your work. Listen back and write two columns: what you like, and what the voice says about you that you did not mean. Repeat monthly and compare against the first.

WATCH FOR

Assuming good content is enough. Correct words said in the wrong voice are not effective, and listeners decide on confidence and trust within seconds.

CHECK YOURSELF · ANSWERS AT THE BACK

17.1 Besides visual image, what makes up the other half of a first impression?

A Your verbal content — the words you choose to say first
B Your vocal image — what people conclude when you speak
C Your body language — what people read in how you stand
D Your credentials — what people know about you beforehand

17.2 What does the 'great career path' demonstration show?

A The same words can mean opposite things, depending on how they're said
B Sarcasm is the hardest tone to read, so avoid it at work
C Praise sounds false unless it names something specific about the person
D Listeners believe the words over the tone when the two disagree

17.3 Why make one video a month?

A To build an archive of clips you can post on social media
B To compare your voice with the speakers you've chosen as models
C To make otherwise invisible progress visible, one marker a month
D To replace live practice, which is hard to schedule every week

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MATH 110 · LINEAR ALGEBRA · LECTURE 18 OF 34

Inverse matrices, column space and null space

Tue 20 Oct · 12 min video · Watch on YouTube

THE TAKEAWAY

Solving $A\vec{x} = \vec{v}$ asks which vector lands on $\vec{v}$, and the determinant decides whether that question has one answer.

THE ARGUMENT · 6 STEPS

1. **A linear system is a transformation question**
   Pack the coefficients into a matrix A and the unknowns into a vector $\vec{x}$. Then $A\vec{x} = \vec{v}$ stops being a tangle of variables and becomes: which input vector lands on $\vec{v}$?

2. **Non-zero determinant: play the tape backwards**
   Space was not squashed, so exactly one vector lands on $\vec{v}$, and you find it by running the transformation in reverse. That reverse is itself a linear transformation — the inverse, A^{-1}.

3. **The inverse is defined by undoing**
   $A^{-1}A$ is the transformation that does nothing: the identity, whose columns are $(1, 0)$ and $(0, 1)$. That single property is the whole content of the word inverse.

4 **Zero determinant: no inverse exists**

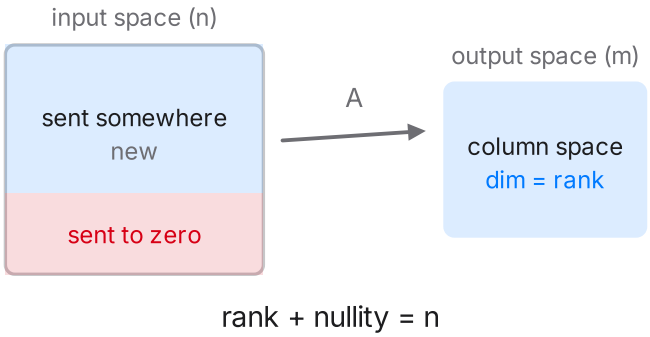
You cannot unsquish a line back into a plane. A function sends one input to one output, and undoing a collapse would require sending one vector to a whole line of them.

5 **Rank names how much survived**

The number of dimensions in the output. Rank 1 is a line, rank 2 a plane. Full rank means the rank equals the number of columns, so nothing collapsed at all.

6 **Column space and null space**

The column space is the span of the columns — everything you can reach. The null space is everything crushed onto the origin, and it is exactly the solution set when $\vec{v}$ is zero.



rank + nullity = n

WORKED EXAMPLE

A collapse is not always fatal. If A squashes the plane onto a line, $A\vec{x} = \vec{v}$ still has solutions — infinitely many — as long as $\vec{v}$ happens to lie on that line. Off it, there are none.

WATCH FOR

Reading $\det A = 0$ as no solution. It means no INVERSE — solutions still exist, a whole line of them, whenever $\vec{v}$ lies in the column space.

KEY CONCEPTS

Rank plus nullity is conserved

CHECK YOURSELF · ANSWERS AT THE BACK

18.1 A 3×3 matrix squashes all of space onto a plane. Its rank is:

A 0
B 2
C 3
D Undefined for a collapse

18.2 A matrix with 7 columns has rank 3. The dimension of its null space is:

Your answer: _____

MATH 110 · LINEAR ALGEBRA · LECTURE 19 OF 34

Nonsquare matrices as transformations between dimensions

Tue 20 Oct · 4 min video · Watch on YouTube

THE TAKEAWAY

The shape of a matrix is not bookkeeping: the columns count the input's dimensions and the rows count the output's.

THE ARGUMENT · 5 STEPS

1. **Transformations between dimensions are ordinary**
   The same two conditions apply — grid lines parallel and evenly spaced, origin fixed. Nothing in the definition requires the input and the output to live in the same space.

2. **Columns are inputs, rows are outputs**
   Three rows and two columns means two basis vectors, each landing somewhere that takes three numbers to describe. So a 3×2 matrix maps two dimensions into three.

3. **Full rank does not mean square**
   That 3×2 map sends everything onto a plane through the origin in 3D. The plane is its column space, and the matrix is full rank because the plane's dimension matches the input's.

4. **Reading the shape is enough**
   A 2×3 matrix: three columns, so you start with three basis vectors in 3D; two rows, so each lands somewhere named by two numbers. A map from space onto the plane, read off without computing.

5. **Down to one dimension is the interesting case**
   A 1×2 matrix takes 2D vectors to single numbers. Each column holds one number — where that basis vector landed on the line. This turns out to be the dot product wearing a different hat.

WORKED EXAMPLE

Take a line of evenly spaced dots in the plane and push them onto the number line. Linear means they stay evenly spaced after landing. The 1×2 matrix that does it is just the two numbers $\hat{\imath}$ and $\hat{\jmath}$ land on.

WATCH FOR

Reading a 3×2 matrix as a square one missing a column. Its input and output live in different spaces entirely — the vectors going in never meet the ones coming out.

KEY CONCEPTS

A matrix moves space

CHECK YOURSELF · ANSWERS AT THE BACK

19.1 A 2×3 matrix describes a transformation:

A From 2D to 3D
B From 3D to 2D
C From 2D to 2D
D Only from a space to itself

19.2 A matrix mapping 4D to 2D has how many columns?

Your answer: _____

MATH 110 · LINEAR ALGEBRA · LECTURE 20 OF 34

Dot products and duality

Tue 20 Oct · 14 min video · Watch on YouTube

THE TAKEAWAY

Every linear map from space down to the number line is secretly a vector, and dotting with that vector is applying the map.

THE ARGUMENT · 5 STEPS

1. **Two definitions that ought to be unrelated**
   Numerically: pair the coordinates, multiply, add. Geometrically: project one vector onto the other's line and multiply the lengths. Positive when they roughly agree, zero when perpendicular, negative when opposed.

2. **Order cannot matter, and symmetry says why**
   With equal lengths, projecting $\vec{w}$ onto $\vec{v}$ is the mirror image of projecting $\vec{v}$ onto $\vec{w}$. Scale one of them and the answer doubles under either reading, so the unequal case agrees too.

3. **Maps to the number line are 1×2 matrices**
   A linear function from 2D to numbers is fixed by where $\hat{\imath}$ and $\hat{\jmath}$ land, and each lands on a single number. Two numbers written in a row — a matrix one row tall.

4. **That matrix is a vector on its side**
   Multiplying a 1×2 matrix by a vector is arithmetically identical to taking a dot product. On its own that is a curiosity; the next beat makes it a theorem.

5. **Duality, proved by a projection**
   Lay a number line diagonally through the origin, with the unit vector $\hat{u}$ sitting at 1. Projecting onto it is linear, so it has a 1×2 matrix — and by symmetry $\hat{\imath}$ lands on $\hat{u}$'s x-coordinate and $\hat{\jmath}$ on its y. The matrix IS $\hat{u}$.

WORKED EXAMPLE

Now scale $\hat{u}$ by 3. Every entry of its matrix triples, so the map becomes project, then multiply by three. That is precisely why $\vec{v} \cdot \vec{w}$ is the projected length times $|\vec{v}|$ rather than the projected length alone.

WATCH FOR

Treating the dot product as a formula that measures angle. It is a linear map, and the vector you dot with is that map drawn as an arrow.

KEY CONCEPTS

Projection finds the closest point

CHECK YOURSELF · ANSWERS AT THE BACK

20.1 The dot product of two perpendicular vectors is zero because:

A They share no coordinates
B The projection of one onto the other is the zero vector
C Their lengths cancel out
D Perpendicular vectors cannot be dotted

20.2 $(1, 2) \cdot (3, 4)$ is:

Your answer: _____

MATH 110 · LINEAR ALGEBRA · LECTURE 21 OF 34

Cross products

Tue 20 Oct · 9 min video · Watch on YouTube

THE TAKEAWAY

The 2D cross product is a signed area; the 3D one is a vector whose length is that area.

THE ARGUMENT · 5 STEPS

1. **In two dimensions it is a signed area**
   Take the parallelogram spanned by $\vec{v}$ and $\vec{w}$. The cross product is its area, positive when $\vec{v}$ lies to the right of $\vec{w}$ and negative when it lies to the left. Order matters; swapping flips the sign.

2. **The basis vectors set the sign convention**
   $\hat{\imath} \times \hat{\jmath}$ is positive, and everything else follows. Since $\hat{\imath}$ sits to the right of $\hat{\jmath}$, that is what fixes right-of as the positive case rather than an arbitrary choice you have to memorise.

3 **Which is exactly a determinant**
Put $\vec{v}$'s coordinates in the first column and $\vec{w}$'s in the second. That matrix sends the unit square to the parallelogram, and the determinant is the factor areas scale by — starting from a square of area 1.

4 **In three dimensions the answer is a vector**
Its length is the area of the parallelogram, and its direction is perpendicular to both. Two directions qualify, so the right-hand rule picks one: forefinger along $\vec{v}$, middle finger along $\vec{w}$, thumb gives the answer.

5 **The computation puts basis vectors in a column**
Write $\hat{\imath}, \hat{\jmath}, \hat{k}$ down the first column, $\vec{v}$ and $\vec{w}$ in the other two, and compute as if they were numbers. Students are told this is a notational trick — the next lecture shows it is not.

WORKED EXAMPLE

$\vec{v} = (-3, 1)$, $\vec{w} = (2, 1)$. The determinant of the matrix with those columns is $(-3)(1) - (2)(1) = -5$. So the parallelogram has area 5, and the sign is negative because $\vec{v}$ lies to the left of $\vec{w}$.

WATCH FOR

Reading the 2D cross product as a vector. In the plane it is one signed number; only the 3D version returns a vector, and only there does the right-hand rule apply.

KEY CONCEPTS

The determinant is a volume factor

CHECK YOURSELF · ANSWERS AT THE BACK

21.1 Swapping the two vectors in a cross product:

A Leaves it unchanged
B Flips its sign
C Squares it
D Makes it zero

21.2 For $\vec{v} = (-3, 1)$ and $\vec{w} = (2, 1)$, the 2D cross product $\vec{v} \times \vec{w}$ is:

Your answer: _____

MATH 110 · LINEAR ALGEBRA · LECTURE 22 OF 34

Cross products in the light of linear transformations

Tue 20 Oct · 13 min video · Watch on YouTube

THE TAKEAWAY

The cross product is the dual vector of a volume function, which is why a determinant computes it and why it comes out perpendicular.

THE ARGUMENT · 5 STEPS

1. **Duality, recalled**
   Any linear map from a space down to the number line is the same as a dot product with one particular vector — its dual vector. Numerically, it is the map's $1 \times n$ matrix turned on its side.

2. **Define a volume function**
   Hold $\vec{v}$ and $\vec{w}$ fixed and let (x, y, z) vary. Put (x, y, z) in the first column and $\vec{v}, \vec{w}$ in the other two; the determinant is the signed volume of the parallelepiped the three of them span.

3. **It is linear, so it has a dual vector**
   Call it $\vec{p}$. By the definition of a dual vector, $\vec{p} \cdot (x, y, z)$ equals that determinant for every input (x, y, z) — no exceptions, because linearity leaves no room for any.

4. **Computationally, $\vec{p}$ is the basis-vector trick**
   Expand the determinant and collect the coefficients of x, y and z. Those three numbers are $\vec{p}$'s coordinates. Writing $\hat{\imath}, \hat{\jmath}, \hat{k}$ into that column is just a way of signalling to read them as coordinates.

5. **Geometrically, $\vec{p}$ must be perpendicular**
   The volume is the base parallelogram's area times the height perpendicular to it. That is precisely a dot product with a vector perpendicular to both $\vec{v}$ and $\vec{w}$ whose length is that area.

WORKED EXAMPLE

Ask what $\vec{p}$ satisfies $\vec{p} \cdot (x, y, z) = \det[(x, y, z) \,|\, \vec{v} \,|\, \vec{w}]$ for every input. Answer it algebraically and you get the basis-vector trick; answer it geometrically and you get perpendicular-with-area-length. One question, so one vector.

WATCH FOR

Accepting the $\hat{\imath}, \hat{\jmath}, \hat{k}$ column as a mnemonic with no content. It is the computational half of a duality argument, and the coefficients it collects ARE the dual vector's coordinates.

KEY CONCEPTS

The determinant is a volume factor

CHECK YOURSELF · ANSWERS AT THE BACK

22.1 The cross product comes out perpendicular to $\vec{v}$ and $\vec{w}$ because:

A. It is defined to be
B. Volume is base area times PERPENDICULAR height, and that is a dot product with a perpendicular vector
C. Determinants are always perpendicular
D. The right-hand rule forces it

22.2 Two perpendicular vectors each of length 2 span a parallelogram of area:

Your answer: _____

SPCH 100 · STORYTELLING · LECTURE 23 OF 34

Think faster, talk smarter — structure under pressure

Tue 20 Oct – Wed 21 Oct · day 1–2 of 2 · 39 min video · Watch on YouTube

THE TAKEAWAY

Abrahams's method for speaking on the spot: be interested rather than interesting, treat anxiety as go-time, take a beat, trade perfection for connection, and reach for a practised structure — Problem-Solution-Benefit, or Answer-Detail-Describe — instead of a list.

THE ARGUMENT · 7 STEPS

1. **Interested, not interesting**
   Start conversations with curiosity: comment on your surroundings ('did I miss the memo on the blue?') and keep them going with 'tell me more'.

2. **Anxiety has symptoms and sources**
   It is hard-wired but manageable. Reframing nerves as excitement helps: it is your body saying this matters.

3. **Three common mistakes**
   Not preparing; believing there is one right way to say it; and answering instantly. Take a beat — a clarifying question or a paraphrase buys the moment you need.

4. **Structure, not lists**
   Our brains are not set up for lists. Problem-Solution-Benefit is the structure of every advertisement and works for any pitch.

5. **Small talk is hacky sack, not tennis**
   Keep the ball in the air together. Mix supporting turns (ask more about their topic) with shifting turns (move the topic) — roughly two-thirds to three-quarters supporting.

6. **Q&A as opportunity, answered with ADD**
   Answer, give a Detailed example, then Describe why it matters to them. It keeps you from taking the listener on your discovery of the answer.

7. **Maximise mediocrity; breathe out**
   Judging yourself mid-sentence steals bandwidth, so turn the judging down. Finish each thought out of breath: the inhale becomes a pause and leaves no room for 'um'.

WORKED EXAMPLE

Asked 'so what do you do?', answer in ADD: one sentence that answers it, one concrete example, then why it matters to the person asking. Then stop.

WATCH FOR

Thinking out loud — listing points as they occur to you. It says far more than needed and leaves the listener with nothing they can remember.

CHECK YOURSELF · ANSWERS AT THE BACK

23.1 According to Abrahams, what is the wrong place to start small talk?

A Asking the other person an open question
B Commenting on something in your surroundings
C Feeling that you must be interesting
D Saying 'tell me more' before they have finished

23.2 What does ADD stand for?

A Answer, give a Detailed example, Describe why it matters
B Ask a question back, Discuss it, then Decide together
C Agree with the point, Deflect, then Defer to the expert
D Address the asker, Define your terms, Deliver the answer

23.3 How does he reduce filler words like 'um'?

A Speaking a little faster, so there is no gap left to fill
B Memorising the opening so the first words come without 'um'
C Pausing deliberately before each new point while holding your breath
D Finishing each thought out of breath, so the inhale becomes a pause

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MATH 120 · SINGLE-VARIABLE CALCULUS · LECTURE 24 OF 34

Visualizing the chain rule and product rule

Wed 21 Oct · 16 min video · Watch on YouTube

THE TAKEAWAY

There are only three ways to combine functions — add, multiply, compose — so three rules cover every expression, however monstrous.

THE ARGUMENT · 6 STEPS

1. **Three combinations, and that is all**
   Subtracting is multiplying by -1 and adding. Dividing is composing with $1/x$ and multiplying. So sum, product and composition are the complete set, and any tangle of functions peels apart into layers of those three.

2. **Sums stack, so their changes stack**
   For $\sin x + x^2$ the height is one bar on top of another. Nudge x and the total change is the change in one plus the change in the other: $df = \cos x\,dx + 2x\,dx$. The derivative of a sum is the sum of the derivatives.

3. **Products are areas, so draw a box**
   Sides $\sin x$ and x^2, both adjustable as x moves. Nudging x adds a thin strip along the bottom, a thin strip along the side, and a negligible corner carrying dx^2.

4 **Which is exactly "left d right, right d left"**
Bottom strip: $\sin x \cdot d(x^2)$. Side strip: $x^2 \cdot d(\sin x)$. Out of context the mnemonic is strange; as two rectangles it is obvious. A constant factor just scales the box, so it passes straight through.

5 **Composition is three number lines**
x on the first, x^2 on the second, $\sin(x^2)$ on the third. A nudge dx produces a nudge dh on the second line, which produces a nudge on the third. Name the middle value h and the bookkeeping gets easy.

6 **So the chain rule is a cancellation that is real**
$\frac{dg}{dx} = \frac{dg}{dh} \cdot \frac{dh}{dx}$. The dh cancelling is not notational sleight of hand — it is literally the nudge on the middle line, measured once going in and once coming out.

WORKED EXAMPLE

Note which function the outer derivative eats. For $\sin(x^2)$ it is $\cos(x^2)$, not $\cos x$ — the inside stays unaltered — and only then do you multiply by $2x$. Writing it dg/dh rather than dg/dx is the reminder.

WATCH FOR

Differentiating the outer function at x instead of at the inner function's value. $\frac{d}{dx}\sin(x^2)$ is $\cos(x^2) \cdot 2x$; $\cos(x) \cdot 2x$ is a different function entirely.

CHECK YOURSELF · ANSWERS AT THE BACK

24.1 The product rule has two terms because:

A Products are commutative
B Nudging x grows the box along two sides, adding one thin rectangle on each
C One term is for the corner
D Derivatives always come in pairs

24.2 By the chain rule, the derivative of $(x^2)^3$ at $x = 1$ is:

Your answer: _____

MATH 120 · SINGLE-VARIABLE CALCULUS · LECTURE 25 OF 34

What's so special about Euler's number e?

Wed 21 Oct · 14 min video · Watch on YouTube

THE TAKEAWAY

Every exponential is proportional to its own derivative, and e is simply the base for which that proportionality constant is 1.

THE ARGUMENT · 6 STEPS

1 **Over a whole day, the rate looks like the population**

A mass doubling daily goes 8 to 16 between day 3 and day 4 — a rate of 8 per day, which is the mass at the start. Tempting, but it compares across a full day, and the derivative asks about vanishing intervals.

2 **The exponent law splits the problem in two**

$2^{t+dt} = 2^t \cdot 2^{dt}$, so the difference quotient factors as $2^t \cdot \dfrac{2^{dt} - 1}{dt}$. Turning addition in the exponent into multiplication outside is what links tiny steps to rates.

3 **And the right-hand factor has no t in it**

Plug small values in and $\dfrac{2^{dt} - 1}{dt}$ settles on about 0.6931 — a constant, independent of where you started. So $\frac{d}{dt}2^t$ is not itself, but is PROPORTIONAL to itself.

4 **Different bases, different constants, and a pattern**

Base 3 gives 1.0986; base 8 gives 2.079, which is exactly three times base 2's constant. Not random. The question becomes: is there a base whose constant is 1?

5 **There is, and that is what defines e**

$e \approx 2.71828$, and e^t equals its own derivative. Asking why e has this property is like asking why π is the circumference-to-diameter ratio — it is not a coincidence, it is the definition.

6 **So the mystery constants are natural logs**

Write $2 = e^{\ln 2}$, so $2^t = e^{(\ln 2)t}$, and the chain rule gives derivative $(\ln 2) \cdot 2^t$. And $\ln 2 = 0.6931$. Every mystery constant is the natural log of the base.

WORKED EXAMPLE

This is why applications write e^{kt} rather than a^t. Population growth, a cup of water cooling, money compounding — all are rates proportional to the quantity itself, and writing them with base e makes k literally that proportionality constant.

WATCH FOR

Thinking e is somehow built into a function written e^{kt}. Any exponential can be written in any base; choosing e is a choice, made because it gives the exponent's constant a readable meaning.

CHECK YOURSELF · ANSWERS AT THE BACK

25.1 $\frac{d}{dt}2^t$ **is proportional to** 2^t **because:**

A 2 is even

B 2^{t+dt} factors as $2^t \cdot 2^{dt}$, so everything depending on dt separates from t entirely

C The graph is a straight line

D Exponentials have no derivative

25.2 **The proportionality constant for base 8 divided by the constant for base 2 is:**

Your answer: _____

MATH 120 · SINGLE-VARIABLE CALCULUS · LECTURE 26 OF 34

Implicit differentiation, what's going on here?

Wed 21 Oct · 16 min video · Watch on YouTube

THE TAKEAWAY

Implicit differentiation stops being strange once you read $2x\,dx + 2y\,dy$ as how much a two-variable expression changes under a step (dx, dy).

THE ARGUMENT · 6 STEPS

1. **An implicit curve has no input and no output**
   $x^2 + y^2 = 5^2$ is the set of points satisfying a property, not the graph of a function. So you cannot nudge "the input" — x and y are interdependent, and the usual derivative story does not apply.

2. **The procedure looks like nonsense the first time**
   Differentiate both sides: x^2 gives $2x\,dx$, y^2 gives $2y\,dy$, the constant gives 0. Rearranged, $dy/dx = -x/y$. It works — but what does differentiating an expression with two variables in it even mean?

3. **The ladder problem is the same equation with time in it**
   A 5 m ladder slipping down a wall: $x(t)^2 + y(t)^2 = 5^2$ holds at every instant, so its time-derivative is 0. That gives $2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0$ — and there the derivative has an obvious meaning.

4. **The circle case is that, with dt removed**
   Name $s = x^2 + y^2$, a function assigning a number to every point of the plane. Then $2x\,dx + 2y\,dy$ is a recipe: given where you are and which tiny step you take, it tells you how much s changes. The nudges float free of any common variable.

5. **Setting the derivative to 0 is the condition for staying on the curve**
   On the circle s is 25 and must remain 25, so $ds = 0$. That is exactly $2x\,dx + 2y\,dy = 0$. Strictly it keeps you on the TANGENT line, but for small enough steps those coincide.

6. **Which turns into a way of deriving new formulas**
   For $y = \ln x$, rewrite as $e^y = x$. A step keeping you on the curve needs $e^y dy = dx$, so $dy/dx = 1/e^y$ — and on the curve e^y IS x. Hence $\frac{d}{dx}\ln x = \frac{1}{x}$.

WORKED EXAMPLE

For $\sin(x)\,y^2 = x$: the left side changes by $\sin x \cdot 2y\,dy + y^2 \cos x\,dx$ (product rule), the right by dx. Setting them equal says both sides must change by the same amount — the only way the equation can keep holding.

WATCH FOR

Reading $ds = 0$ as "nothing changes". The step is real and both x and y move; what stays fixed is the VALUE of the expression, which is what keeps you on the curve rather than drifting off it.

CHECK YOURSELF · ANSWERS AT THE BACK

26.1 Setting the differentiated expression equal to zero means:

A The curve is flat
B The step (dx, dy) keeps the expression's value unchanged, so it stays on the curve
C x and y are both zero
D The derivative does not exist

26.2 On the circle $x^2 + y^2 = 25$, the slope dy/dx at the point $(3, 4)$ is:

Your answer: _____

MATH 130 · PROBABILITY · LECTURE 27 OF 34

Story Proofs, Axioms of Probability

Thu 22 Oct · 46 min video · Watch on YouTube

THE TAKEAWAY

A story proof counts the same thing two ways and reads off the identity — and the whole of probability rests on just two axioms once you stop assuming outcomes are equally likely.

THE ARGUMENT · 12 STEPS

1. **Label everything**
Given n people, n elk or n balls, imagine them numbered 1 to n. Ten green balls may look identical to you, but probability behaves as though they are distinguishable, and reasoning as if they are not will give wrong answers.

2. **Check answers by extreme and simple cases**
Looking over your work again finds nothing, because you will make the same mistake twice. Plug in $k = 0$, $k = 1$, $n = 2$ — the simplest NONTRIVIAL case is usually the informative one.

3. **Teams of 4 and 6 versus two teams of 5**
$\binom{10}{4}$ splits into a 4 and a 6, because naming the four names the six. But two teams of 5 is $\binom{10}{5}/2$: with no jerseys to tell them apart, choosing $\{1..5\}$ and choosing $\{6..10\}$ are the same split counted twice.

4. **Which also proves $\binom{n}{k} = \binom{n}{n-k}$**
Choose who is in or choose who is out. One sentence, no factorials — that is a story proof: proof by interpretation rather than by algebra.

5. **Stars and bars**
Choosing k from n with replacement, order not mattering, is the same problem as putting k INDISTINGUISHABLE particles into n distinguishable boxes. Encode a configuration as k dots and $n - 1$ separators; choose which of the $n + k - 1$ positions hold the dots. Hence $\binom{n+k-1}{k}$.

6 **Recognising two problems are the same is the hard skill**

Nothing in the original wording mentions particles or boxes. Seeing that the dot diagram already IS the answer is pattern recognition, and it is what the course is training.

7 **Bose, Einstein, and when indistinguishability is real**

For two coins we count four outcomes even when the coins look identical, because they are really labelled. Bose proposed that certain particles have only THREE — no distinction between HT and TH — was laughed at, wrote to Einstein, and the two predicted a state of matter observed seventy years later. For probability, the labelled model is almost always the right one.

8 **The committee-with-a-president identity**

$k\binom{n}{k} = n\binom{n-1}{k-1}$. Choose the committee then its chair, or choose the chair then the rest of the committee from those remaining. Both count the same configurations, so they are equal.

9 **Vandermonde, which algebra makes horrible**

$\binom{m+n}{k} = \sum_{j=0}^{k} \binom{m}{j}\binom{n}{k-j}$. Split the $m+n$ people into a group of m and a group of n; any choice of k takes j from the first and $k-j$ from the second, and the cases are disjoint. Four lines instead of a page of cancelling factorials.

10 **A probability space is a pair**

S, the sample space, and P, a function whose INPUT is an event — a subset of S — and whose output is a number in $[0,1]$. Note that P does not take outcomes; it takes sets.

11 **Axiom 1: the two extremes**

$P(\varnothing) = 0$ and $P(S) = 1$. An event occurs when the realised outcome lies in it; nothing lies in the empty set, and everything lies in S.

12 **Axiom 2: countable additivity**

$P\left(\bigcup_n A_n\right) = \sum_n P(A_n)$ when the A_n are DISJOINT. That condition is the whole content. Every theorem in probability follows from these two rules.

WORKED EXAMPLE

When an identity looks like a nightmare in algebra, read both sides as counting instructions. $\binom{m+n}{k}$ says 'choose k from two groups combined'; the sum says 'choose j from one and the rest from the other'. If both descriptions count the same configurations, the identity is proved.

WATCH FOR

Dividing by 2 out of habit, or failing to. The question is whether anything distinguishes the two groups: teams of 4 and 6 differ by size, two teams of 5 do not. Ask what makes two outcomes different before deciding whether you have double-counted.

CHECK YOURSELF · ANSWERS AT THE BACK

27.1 Why is splitting 10 people into two teams of 5 counted as $\binom{10}{5}/2$?

A Because order matters within a team

B Because nothing distinguishes the teams, so each split is counted once as 'this five' and once as 'the other five'

C Because $\binom{10}{5}$ is even

D It should not be divided; the answer is $\binom{10}{5}$

27.2 The number of ways to split 10 people into two unlabelled teams of 5 is:

Your answer: _____

27.3 $\sum_{j=0}^{5} \binom{3}{j}\binom{5}{5-j}$ equals:

Your answer: _____

AI 200 · NEURAL NETWORKS: ZERO TO HERO · LECTURE 28 OF 34

Makemore 1 — bigram character-level language model

Thu 22 Oct – Fri 23 Oct · day 1–2 of 3 · 118 min video · Watch on YouTube · finishes in week 2

THE TAKEAWAY

A bigram language model can be built two ways — count which character follows which and normalise, or train a one-layer net with a softmax on the negative log likelihood — and the two arrive at the same model, which is the point: the neural framing is the one that scales.

THE ARGUMENT · 8 STEPS

1. **One file of names, many examples**
   32,033 names, 2 to 15 letters long. Each name is several examples at once: every character is evidence about which character follows it, and where names start and end.

2. **Counting bigrams**
   Wrap each name in start and end markers and count every adjacent pair. The final version uses one token, '.', at index 0 for both start and end, giving a 27 × 27 table N whose row i counts what follows character i.

3. **From counts to samples**
   Normalise each row into probabilities, start from '.', and draw the next character with torch.multinomial until '.' comes back. The names ('mor.', 'axx.', 'minaymoryles.') are bad — a bigram sees one character of context — but they are drawn from the data's statistics.

4 **Broadcasting, carefully**

Normalise with P.sum(1, keepdims=True): a 27 × 1 column of row sums divides each ROW. P.sum(1) gives shape 27, which broadcasts as a 1 × 27 row and divides each COLUMN instead — no error, just a wrong model.

5 **Scoring a model: negative log likelihood**

Maximising the likelihood of the data — the product of the probabilities assigned to every actual next character — is the same as minimising the average negative log likelihood. The smoothed count model scores 2.4765. Adding 1 to every count keeps any pair from having probability 0, which would make the loss infinite.

6 **The same model as a neural net**

One-hot encode the input character and multiply by a 27 × 27 weight matrix W: that selects W's row for the character — 27 logits, read as log-counts. Exponentiate and normalise (together, a softmax) to get probabilities. The loss is the mean NLL of the correct next characters.

7 **Training it**

On all 228,146 bigrams: forward, set W.grad to None, loss.backward(), W += -50 * W.grad. A regulariser, 0.01 * (W**2).mean(), pulls W toward zero, where all logits are equal and predictions are uniform — the gradient-based twin of adding counts.

8 **Same destination**

The trained net reaches a loss of about 2.48, the counts' level, and with the same random seed it samples almost exactly the same names. The difference is what comes next: the table cannot grow to longer contexts, and the net can.

WORKED EXAMPLE

Index the probabilities of the right answers in one step — probs[torch.arange(n), ys] — take the log, average and negate: that is the loss, for the count table and the network alike. Then compare it against the counts model's 2.48 as the target a one-character-context model can reach.

WATCH FOR

Dropping keepdims=True when normalising. P.sum(1) has shape 27 and broadcasts as a row, so each column is divided by a row's total. Nothing errors and the samples still look plausible; only the numbers are wrong.

CHECK YOURSELF · ANSWERS AT THE BACK

28.1 The first count table marks names with separate start and end tokens; the final one uses a single '.' at index 0 for both. Why is one token enough?

A Names never contain a dot, so the model cannot mistake it for a letter
B A start only ever opens a bigram and an end only closes one, so they never clash
C PyTorch needs the count table to be square, so the two markers had to be merged
D One shared token means every sampled name starts and ends with one letter

28.2 How many training bigrams does the name 'isabella' contribute, once it is wrapped in the start/end token?

Your answer: _____

28.3 P is 27 × 27. Why must each row be divided by P.sum(1, keepdims=True) rather than by P.sum(1)?

A keepdims makes the sum run over columns instead of rows, as is needed
B Without keepdims PyTorch raises an error, because the shapes cannot broadcast
C The two are the same; keepdims only changes how the result is printed
D Without keepdims the sums broadcast as a row, and columns get divided

SPCH 100 · STORYTELLING · LECTURE 29 OF 34

The Pixar story spine — six prompts that hold a story up

Thu 22 Oct · 4 min video · Watch on YouTube

THE TAKEAWAY

The story spine reduces any story to six prompts — Once upon a time, Every day, One day, Because of that, Because of that, Until finally — a wireframe to build detail onto, and it works on a Pixar film, on history, or on a trip to the grocery store.

THE ARGUMENT · 5 STEPS

1. **A wireframe first**
   Distil the story to its most basic form — a gesture if you draw, an outline if you write — so that every scene you add later pushes the plot forward.

2. **The six prompts**
   Once upon a time... Every day... One day... Because of that... Because of that... Until finally...

3. **Up**
   A couple saves for adventure; one day she dies; because of that he flies off, takes a scout along, compromises; until finally he learns the ultimate adventure is caring for those you love.

4 **Dead Poets Society and Martin Luther**

The same six prompts carry a school drama and a religious revolution — the structure is scale-free.

5 **Know where it ends**

Once you know where you start and where the story ends, it is far easier to choose details that support the ending. Try it on a trip to the grocery store.

WORKED EXAMPLE

Fill all six prompts in one sitting — ending first if that is easier — then read the two 'because of that' lines aloud and check that each is caused by the line before it.

WATCH FOR

Letting 'because of that' quietly become 'and then'. A list of events in order is not a chain of causes, and it is the causes that push a story forward.

CHECK YOURSELF · ANSWERS AT THE BACK

29.1 **Which prompt forces cause and effect?**

A Once upon a time
B Every day
C Because of that
D Until finally

29.2 **What does the spine give you before any details?**

A A wireframe or outline of the whole story
B A cast of characters and what each one wants
C The final line, so you know where to stop
D A title and a theme to build the scenes on

29.3 **What ordinary story does the video suggest practising on?**

A A film you watched last week
B Your CV, told as a career story
C A family wedding
D Going to the grocery store

SPCH 100 · STORYTELLING · LECTURE 30 OF 34

How to easily be funnier in conversations

Thu 22 Oct · 5 min video · Watch on YouTube

THE TAKEAWAY

Most conversational humour is prepared: a true answer and a playful answer to the questions you are always asked, saying the opposite as a cheap and reliable move, and priming yourself with ten minutes of a comedian you love. The deep path is an improv class.

THE ARGUMENT · 5 STEPS

1. **The same questions, every time**
   Conversations open on the same tracks — what do you study, what do you do. Prepare a true answer that shows your values and a playful one, and split-test them, as he did.

2. **Say the opposite**
   On a hot day when everyone is complaining, 'man, it's cold out here' gets a chuckle. Much of humour is simply the unexpected.

3. **Prime yourself**
   Watch ten to fifteen minutes of the comedian whose vibe you want before you go out. You carry their demeanour in, not their lines.

4. **The bar is low**
   You do not need to be excellent. Saying the opposite will make you the funniest person in most groups.

5. **Go deeper with improv**
   Improv classes teach scene-building and structure — almost everything about being funny live, if the class is any good.

WORKED EXAMPLE

Write your three most-asked questions and, under each, a true answer and a playful one. Use the playful ones in real conversations and keep whichever actually gets a laugh.

WATCH FOR

Trying to be spontaneously quick like a professional comedian with no prepared material. Speed comes from long practice; the prepared answer is available today.

CHECK YOURSELF · ANSWERS AT THE BACK

30.1 What does he suggest preparing for the questions you are asked most?

A A true answer that reveals your values, and a playful one
B A memorised joke that works for any version of the question
C A question to ask back, so the focus moves off you
D A short story about the most surprising answer you've had

30.2 Why does saying the opposite tend to get a chuckle?

A People enjoy anyone who disagrees with the group
B Humour is often just the unexpected happening
C It shows you're confident enough to be wrong
D It changes the subject away from a complaint

30.3 What does 'priming yourself' mean here?

A Rehearsing three jokes out loud before you walk into the room
B Reading the news first so you have topical material to riff on
C Watching ten minutes of a comedian whose vibe you want, before you go
D Borrowing a comedian's best lines and using them in conversation later

MATH 110 · LINEAR ALGEBRA · LECTURE 31 OF 34

Cramer's rule, explained geometrically

Fri 23 Oct · 12 min video · Watch on YouTube

THE TAKEAWAY

Each coordinate of the unknown vector is an area, and every area scales by the determinant — that is the whole of Cramer's rule.

THE ARGUMENT · 5 STEPS

1. **A wrong idea, pointing the right way**
   You might hope that dotting the output with the transformed basis vectors recovers the coordinates. It does not: transformations rarely preserve dot products. The ones that do are called orthonormal, and for those the hope is exactly right.

2. **A coordinate, written as an area**
   The parallelogram spanned by $\hat{\imath}$ and the unknown vector has base 1, so its signed area IS that vector's y-coordinate. Pairing the unknown vector with $\hat{\jmath}$ gives the x-coordinate the same way.

3 **In three dimensions it is a volume**
Pair the unknown vector with the two basis vectors you are not asking about. The signed volume of that parallelepiped is the coordinate you want, and the right-hand rule keeps the sign honest.

4 **Areas do not survive, but they all scale alike**
That is what the determinant is. After the transformation every parallelogram's area has been multiplied by $\det A$ — the same factor for all of them — so dividing by it undoes the damage.

5 **Which makes it computable from the output alone**
Replace one column of A with the output vector and take that determinant, then divide by $\det A$. Replacing the first column gives x, the second gives y. Nothing but numbers you were handed.

WORKED EXAMPLE

$A = \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}$, $\vec{v} = (9, 8)$. Swap the output into column one: $\det \begin{pmatrix} 9 & 1 \\ 8 & 2 \end{pmatrix} = 10$, and $\det A = 5$. So $x = 10/5 = 2$, and back-substituting gives $y = 3$.

WATCH FOR

Reaching for Cramer's rule to actually solve systems. Gaussian elimination is always faster; this exists to show the determinant and the solution are one fact.

KEY CONCEPTS

The determinant is a volume factor

CHECK YOURSELF · ANSWERS AT THE BACK

31.1 Cramer's rule works because a linear transformation:

A Preserves all areas
B Scales every area by the same factor, its determinant
C Preserves dot products
D Keeps the basis vectors perpendicular

31.2 For $A = \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}$ and $\vec{v} = (9, 8)$, Cramer's rule gives $x =$

Your answer: _____

MATH 110 · LINEAR ALGEBRA · LECTURE 32 OF 34

Change of basis

Fri 23 Oct · 13 min video · Watch on YouTube

THE TAKEAWAY

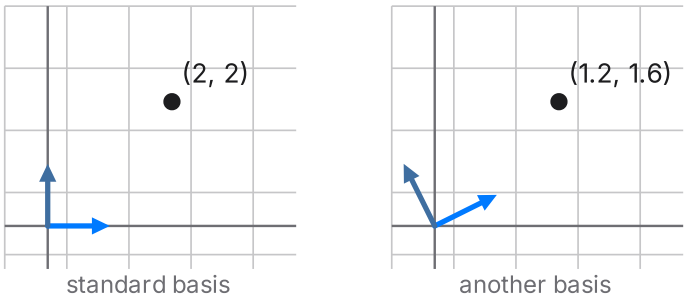
A matrix whose columns are someone else's basis vectors translates their coordinates into yours; its inverse goes the other way.

THE ARGUMENT · 5 STEPS

1. **Coordinates are a language, not the vector**
   $(3, 2)$ means scale $\hat{\imath}$ by 3, scale $\hat{\jmath}$ by 2, add. Those two vectors carry every implicit assumption: which way is right, which is up, how long one unit is.

2. **Another basis is another language**
   Jennifer's $\vec{b}_1$ and $\vec{b}_2$ are $(2, 1)$ and $(-1, 1)$ in our coordinates. In hers they are $(1, 0)$ and $(0, 1)$ — they are what those coordinates MEAN in her world. Same arrows in space, different words.

3. **The grid is a construct**
   Space has no grid of its own; a grid is a drawing of a chosen basis. Only the origin is agreed on, because everybody's $(0, 0)$ is what you get by scaling anything by zero.



standard basis — another basis

4. **Her language into ours is a matrix-vector product**
   Put her basis vectors, written in our coordinates, into the columns. Multiplying by her coordinates scales those columns and adds them — which is exactly what her coordinates were instructing all along.

5. **A transformation from her seat is $A^{-1}MA$**
   Translate her vector into our language, apply the transformation there, translate the result back. The middle matrix is the transformation as we see it; the outer two are the change of perspective.

WORKED EXAMPLE

The vector she calls $(-1, 2)$: compute $-1 \cdot (2, 1) + 2 \cdot (-1, 1) = (-4, 1)$, and that is it in our coordinates. Going the other way, our $(3, 2)$ is her $(5/3,\ 1/3)$ — same arrow, inverse matrix.

WATCH FOR

Expecting the matrix to convert OUR coordinates into hers, because geometrically it moves our grid onto hers. Numerically it does the exact opposite.

KEY CONCEPTS

A basis is the ruler you measure with

CHECK YOURSELF · ANSWERS AT THE BACK

32.1 The expression $A^{-1}MA$ represents:

A The transformation M applied twice
B The same transformation M, described in the other basis
C The inverse of M
D A rotation composed with M

32.2 Her basis vectors are $(2, 1)$ and $(-1, 1)$ in our coordinates. The vector she calls $(-1, 2)$ has our first coordinate:

Your answer: _____

MATH 110 · LINEAR ALGEBRA · LECTURE 33 OF 34

Eigenvectors and eigenvalues

Fri 23 Oct · 17 min video · Watch on YouTube

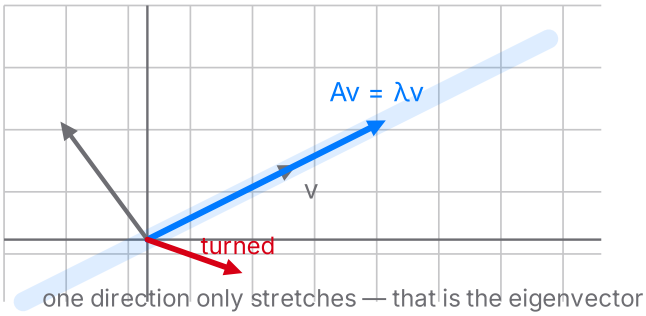
THE TAKEAWAY

An eigenvector is a direction the transformation only stretches, never turns, and its eigenvalue is that stretch.

THE ARGUMENT · 6 STEPS

1 **Most vectors get knocked off their line**

Apply a transformation and watch a vector: usually it ends up pointing somewhere new. Some special directions stay on their own line and are merely scaled. Those are the eigenvectors.



2 **The defining equation**

$Av = \lambda v$ with $v \neq 0$. The matrix on the left and the plain number on the right do the same thing to that one vector — which is why eigenvectors make matrix powers easy.

3 **Finding them means engineering a collapse**

Rewrite as $(A - \lambda I)v = 0$. A nonzero v can only be sent to zero if $A - \lambda I$ squashes space, so $\det(A - \lambda I) = 0$. That is the characteristic equation, and this is where the determinant earns its keep.

4 **Then solve for the direction**

Each root λ gives a matrix $A - \lambda I$ whose null space is the eigenvectors for that λ — a whole line of them, since scaling an eigenvector leaves it an eigenvector.

5 **Not every map has enough of them**

A shear has a single eigendirection; some rotations have none at all in the real plane, because every direction genuinely turns. An eigenbasis is a privilege, not a guarantee.

6 **Why anyone cares**

In an eigenbasis the transformation is just independent stretches along axes, so applying it a hundred times means raising numbers to the hundredth power instead of multiplying matrices.

WORKED EXAMPLE

$A = \begin{pmatrix} 3 & 1 \\ 0 & 2 \end{pmatrix}$. Then $\det(A - \lambda I) = (3 - \lambda)(2 - \lambda)$, giving $\lambda = 3, 2$ — the diagonal, as always for a triangular matrix. For $\lambda = 3$, solve $(A - 3I)v = 0$: $\begin{pmatrix} 0 & 1 \\ 0 & -1 \end{pmatrix} v = 0$ forces the second component to zero, so $v = (1, 0)$.

WATCH FOR

Assuming every matrix has a full set of eigenvectors. Ask what a shear does before you assume an eigenbasis exists.

KEY CONCEPTS

Eigenvectors are the directions that survive

CHECK YOURSELF · ANSWERS AT THE BACK

33.1 The characteristic equation is $\det(A - \lambda I) = 0$ because:

A Determinants are easy to compute
B A nonzero vector can only be sent to zero if the map collapses space
C Every matrix is invertible
D λ must be real

33.2 The largest eigenvalue of $\begin{pmatrix} 3 & 1 \\ 0 & 2 \end{pmatrix}$ is:

Your answer: _____

MATH 110 · LINEAR ALGEBRA · LECTURE 34 OF 34

A quick trick for computing eigenvalues

Fri 23 Oct · 13 min video · Watch on YouTube

THE TAKEAWAY

The trace and the determinant hand you the mean and the product of the eigenvalues, and two numbers are pinned down by those.

THE ARGUMENT · 5 STEPS

1. **Fact one: the trace is the sum**
   The two diagonal entries add to the sum of the eigenvalues, so the mean of the eigenvalues is the mean of the diagonal — and you can read that off the matrix without writing anything down.

2. **Fact two: the determinant is the product**
   Which ought to feel right: an eigenvalue says how much one direction is stretched, and the determinant says how much area is scaled overall. Multiply the stretches and you have the area factor.

3. **Fact three: mean and product pin down the pair**
   Two numbers sitting at $m \pm d$ multiply to $m^2 - d^2$, a difference of squares. Set that equal to the product p and $d = \sqrt{m^2 - p}$ falls straight out.

4. **The whole trick in one line**
   $\lambda = m \pm \sqrt{m^2 - p}$, with m the mean of the diagonal and p the determinant. No characteristic polynomial, no expanding, no quadratic formula.

5 Why it is faster, and when it is not

It is the same mathematics as finding the roots of the characteristic polynomial — it has to be. The saving is that m and p are legible straight off the matrix, so you begin writing the answer immediately.

WORKED EXAMPLE

$\begin{pmatrix} 3 & 1 \\ 4 & 1 \end{pmatrix}$: the mean of 3 and 1 is 2, and the determinant is $3 - 4 = -1$. So the eigenvalues are $2 \pm \sqrt{4 + 1} = 2 \pm \sqrt{5}$. You never wrote down a polynomial.

WATCH FOR

Filing it as one more formula to memorise. Its value is that the trace and the determinant already mean something, so it carries its own derivation.

KEY CONCEPTS

Eigenvectors are the directions that survive

CHECK YOURSELF · ANSWERS AT THE BACK

34.1 In $\lambda = m \pm \sqrt{m^2 - p}$, the quantity p is:

A The trace
B The determinant
C The mean of the diagonal
D The larger eigenvalue

34.2 For $\begin{pmatrix} 2 & 7 \\ 1 & 8 \end{pmatrix}$, the larger eigenvalue is:

Your answer: _____

WEEK 1 · FRI 16 – FRI 23 OCTOBER 2026

Not yet summarised

4 lectures this week have no summary yet. Watch or rewatch them from the links below.

LECTURES

7 But what is a neural network?
AI 200 · Fri 16 Oct · 19 min video
https://www.youtube.com/watch?v=aircAruvnKk

8 Gradient descent — how neural networks learn
AI 200 · Fri 16 Oct · 21 min video
https://www.youtube.com/watch?v=IHZwWFHWa-w

12 What is backpropagation really doing?
AI 200 · Sun 18 Oct · 13 min video
https://www.youtube.com/watch?v=Ilg3gGewQ5U

13 Backpropagation calculus
AI 200 · Sun 18 Oct · 10 min video
https://www.youtube.com/watch?v=tIeHLnjs5U8

WEEK 1 · FRI 16 – FRI 23 OCTOBER 2026

What this week established

Each lecture’s takeaway, by course.

MATH 110 · Linear Algebra

1 **Vectors.** Everything in linear algebra is built from exactly two operations: adding vectors and scaling them.

2 **Linear combinations, span, and basis vectors.** Coordinates are scalars applied to a chosen basis, and the span is everything those scalars can reach.

3 **Linear transformations and matrices.** A linear transformation is fully determined by where the basis vectors land — and that is all a matrix records.

4 **Matrix multiplication as composition.** Matrix multiplication is doing one transformation after another, which is why order matters and why the rows-times-columns rule looks the way it does.

5 **Three-dimensional linear transformations.** Nothing changes in three dimensions: the transformation is still completely described by where the basis vectors land.

6 **The determinant.** The determinant is the factor by which a transformation scales area, and its sign says whether space was flipped.

18 **Inverse matrices, column space and null space.** Solving $A\vec{x} = \vec{v}$ asks which vector lands on $\vec{v}$, and the determinant decides whether that question has one answer.

19 **Nonsquare matrices as transformations between dimensions.** The shape of a matrix is not bookkeeping: the columns count the input's dimensions and the rows count the output's.

20 **Dot products and duality.** Every linear map from space down to the number line is secretly a vector, and dotting with that vector is applying the map.

21 **Cross products.** The 2D cross product is a signed area; the 3D one is a vector whose length is that area.

22 **Cross products in the light of linear transformations.** The cross product is the dual vector of a volume function, which is why a determinant computes it and why it comes out perpendicular.

31 **Cramer's rule, explained geometrically.** Each coordinate of the unknown vector is an area, and every area scales by the determinant — that is the whole of Cramer's rule.

32 **Change of basis.** A matrix whose columns are someone else's basis vectors translates their coordinates into yours; its inverse goes the other way.

33 **Eigenvectors and eigenvalues.** An eigenvector is a direction the transformation only stretches, never turns, and its eigenvalue is that stretch.

34 **A quick trick for computing eigenvalues.** The trace and the determinant hand you the mean and the product of the eigenvalues, and two numbers are pinned down by those.

AI 200 · Neural Networks: Zero to Hero

7 **But what is a neural network?** Not yet summarised.

8 **Gradient descent — how neural networks learn.** Not yet summarised.

12 **What is backpropagation really doing?** Not yet summarised.

13 **Backpropagation calculus.** Not yet summarised.

16 **Micrograd — spelled-out intro to neural networks and backpropagation.** Backpropagation is the chain rule applied node by node over an expression graph: each operation knows only its local derivative, gradients flow from the output back through a topological order, and they ADD wherever a value is used more than once — about a hundred lines of Python, and the same numbers PyTorch gives.

28 **Makemore 1 — bigram character-level language model.** A bigram language model can be built two ways — count which character follows which and normalise, or train a one-layer net with a softmax on the negative log likelihood — and the two arrive at the same model, which is the point: the neural framing is the one that scales.

MATH 120 · Single-Variable Calculus

9 **The essence of calculus.** Hard problems become sums of many small quantities, those sums become areas under graphs, and derivatives are how you find those areas.

10 **The paradox of the derivative.** "Instantaneous rate of change" is an oxymoron, and the derivative is the clever construction that gets the meaning without the contradiction.

11 **Derivative formulas through geometry.** Derivative formulas are not rules to memorise: draw what the function literally means, nudge the input, and read the answer off the picture.

24 **Visualizing the chain rule and product rule.** There are only three ways to combine functions — add, multiply, compose — so three rules cover every expression, however monstrous.

25 **What's so special about Euler's number e?** Every exponential is proportional to its own derivative, and e is simply the base for which that proportionality constant is 1.

26 **Implicit differentiation, what's going on here?** Implicit differentiation stops being strange once you read $2x\,dx + 2y\,dy$ as how much a two-variable expression changes under a step (dx, dy).

SPCH 100 · Storytelling

14 **Homework for Life — finding the story in an ordinary day.** Homework for Life: at the end of each day, ask what the most story-like moment was and write a line or two. Done daily it builds a storyteller's lens, fills a bank of small stories, and — the part Dicks did not expect — slows time down.

17 **Five ways to improve your professional voice.** Half of a first impression is vocal, and most professionals switch into a flat 'professional' voice. Giang's five fixes: respect what the voice does, build a taste for voices, look in the vocal mirror, read children's books aloud, and record yourself regularly.

23 **Think faster, talk smarter — structure under pressure.** Abrahams's method for speaking on the spot: be interested rather than interesting, treat anxiety as go-time, take a beat, trade perfection for connection, and reach for a practised structure — Problem-Solution-Benefit, or Answer-Detail-Describe — instead of a list.

29 **The Pixar story spine — six prompts that hold a story up.** The story spine reduces any story to six prompts — Once upon a time, Every day, One day, Because of that, Because of that, Until finally — a wireframe to build detail onto, and it works on a Pixar film, on history, or on a trip to the grocery store.

30 **How to easily be funnier in conversations.** Most conversational humour is prepared: a true answer and a playful answer to the questions you are always asked, saying the opposite as a cheap and reliable move, and priming yourself with ten minutes of a comedian you love. The deep path is an improv class.

MATH 130 · Probability

15 **Probability and Counting.** The naive definition — favourable outcomes over possible outcomes — is correct only when the outcomes are equally likely and finite in number, and the rest of this lecture is learning to count the two numbers.

27 **Story Proofs, Axioms of Probability.** A story proof counts the same thing two ways and reads off the identity — and the whole of probability rests on just two axioms once you stop assuming outcomes are equally likely.

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Answers

The answers and explanations are the summaries' own.

1 Vectors

1.1 B — Walking one displacement then the other adds each direction separately

Tip-to-tail movement in each axis accumulates independently, which is exactly componentwise addition.

1.2 4

$1 + 3 = 4$.

2 Linear combinations, span, and basis vectors

2.1 B — A line through the origin

The second vector adds no new direction, so the reachable set stays one line.

2.2 3

Exactly the dimension: enough to span, few enough to stay independent.

3 Linear transformations and matrices

3.1 B — Where the basis vectors land

Grid lines stay parallel and evenly spaced, so the basis images fix everything else.

3.2 1

$\hat{\imath} = (1, 0)$ lands on $(0, 1)$, so the first column is $(0, 1)$ and its second component is 1.

4 Matrix multiplication as composition

4.1 B — B

$(AB)x = A(Bx)$ — the right-hand map acts first, as in $f(g(x))$.

4.2 B — Both describe the same sequence of actions

Associativity is a statement about doing C, then B, then A either way.

5 Three-dimensional linear transformations

5.1 B — Three basis vectors each need three coordinates to say where they landed

Columns count basis vectors, rows count the coordinates describing each landing spot. 3×3.

5.2 −1

The first column is where $\hat{\imath}$ lands, and it lands one unit down the negative z-axis.

6 The determinant

6.1 B — Collapses space into a lower dimension

Zero area factor means the image is degenerate, so the columns are dependent and A is not invertible.

6.2 6

A 3-by-2 rectangle: area $3 \cdot 2 = 6$.

9 The essence of calculus

9.1 B — Each term is a rectangle of width dr and height $2\pi r$, and they pack side by side under $y = 2\pi r$
The factor dr is both the thickness of each rectangle and the spacing between successive values of r, which is what lets them sit flush against each other beneath the graph.

9.2 9
The sliver added by nudging x is a rectangle of height x^2 and width dx, so $dA/dx = x^2 = 3^2 = 9$. That relation IS the fundamental theorem of calculus.

10 The paradox of the derivative

10.1 B — It is what that ratio APPROACHES as dt shrinks — a limit, not a value at any particular dt
Any fixed dt gives the slope of a secant through two points. As $dt \to 0$ that secant approaches the tangent at a single point, and the derivative is the tangent's slope — never evaluated AT $dt = 0$, which would be division by zero.

10.2 12
Expanding $(2 + dt)^3 - 2^3$ and dividing by dt leaves $3 \cdot 2^2$ plus terms that all carry a dt. Those vanish, giving $3 \cdot 4 = 12$.

11 Derivative formulas through geometry

11.1 B — The nudge triangle on the unit circle is similar to the angle-θ triangle, so the ratio IS adjacent over hypotenuse
Reading the graph gives the shape and nothing more — many functions share those peaks. The similar triangles turn $d(\sin\theta)/d\theta$ into a ratio that is the definition of cosine.

11.2 80
$\frac{d}{dx}x^5 = 5x^4$, so at $x = 2$ it is $5 \cdot 16 = 80$. The 5 counts the parentheses from which the single dx could have been chosen.

14 Homework for Life — finding the story in an ordinary day

14.1 B — What was the most story-like moment of today, if I had to tell a story from it?
The words 'as benign as it might be' matter: the point is the best moment of an ordinary day.

14.2 A — It would be too much work, and he would stop doing it
He says outright that writing the story would be too much. A sentence or two keeps the habit alive.

14.3 C — Time slowing down, so days stop disappearing
He calls it a magical elixir: the days creep by, and none is lost.

15 Probability and Counting

15.1 B — Because listing two outcomes does not make them equally likely, and the naive definition requires that

The same argument gives $\frac{1}{2}$ for intelligent life, which must be strictly less probable than life. The contradiction exposes the missing justification rather than an arithmetic slip.

15.2 0.00144

$\frac{13\binom{4}{3}\cdot 12\binom{4}{2}}{\binom{52}{5}} = \frac{3744}{2598960} = 0.001441$. The second rank has 12 choices, not 13 — it cannot repeat the first.

15.3 220

$\binom{n+k-1}{k} = \binom{12}{3} = 220$ — the one cell of the sampling table that does not fall out of the multiplication rule. Compare $10^3 = 1000$ when order does matter.

16 Micrograd — spelled-out intro to neural networks and backpropagation

16.1 14

The nudge estimates the derivative $f'(x) = 6x - 4$, so at $x = 3$ the slope is 14: raise x by h and f rises by about 14h. At $x = 2/3$ the same estimate gives about 0 — the bottom of the parabola.

16.2 A — g to each input, unchanged, since a sum's local derivative is 1

Nudging c in d = a*b + c moves d by exactly h, so the local derivative is 1, and the chain rule multiplies g by 1 for both inputs — a + node routes the gradient through to everything that feeds it. Multiplying by the other input's value is the rule for *.

16.3 −4

Chain rule: dL/dd = f = -2, dd/de = 1, and de/db = a = 2, so dL/db = -2 × 1 × 2 = -4. The notebook's lol() checks it numerically by nudging b and gets -3.99999.

17 Five ways to improve your professional voice

17.1 B — Your vocal image — what people conclude when you speak

He splits the first impression roughly fifty-fifty between visual image and vocal image.

17.2 A — The same words can mean opposite things, depending on how they're said

The same sentence came out amazed, sarcastic, defeated and disappointed.

17.3 C — To make otherwise invisible progress visible, one marker a month

Progress in communication is hard to see; twelve markers a year make it visible and keep you going.

18 Inverse matrices, column space and null space

18.1 B — 2

Rank is the dimension of the output. A plane is two-dimensional, so the rank is 2 — collapsed, but not as far as a line would be.

18.2 4

Rank plus nullity is the number of columns: $7 - 3 = 4$. Four dimensions' worth of input is crushed onto the origin.

19 Nonsquare matrices as transformations between dimensions

19.1 B — From 3D to 2D

Three columns means three input basis vectors, so you start in 3D; two rows means each landing spot needs two coordinates, so you finish in 2D.

19.2 4

One column per input basis vector, and the input space is four-dimensional.

20 Dot products and duality

20.1 B — The projection of one onto the other is the zero vector

The dot product is the projected length times the length projected onto. A perpendicular vector projects to nothing, so the first factor is zero.

20.2 11

$1 \cdot 3 + 2 \cdot 4 = 3 + 8 = 11$.

21 Cross products

21.1 B — Flips its sign

The area is the same, but the orientation reverses — and orientation is what the sign records.

21.2 −5

$\det\begin{pmatrix} -3 & 2 \\ 1 & 1 \end{pmatrix} = -3 - 2 = -5$. Area 5, negative orientation.

22 Cross products in the light of linear transformations

22.1 B — Volume is base area times PERPENDICULAR height, and that is a dot product with a perpendicular vector

The dual vector has to reproduce the volume through a dot product, and only the perpendicular component of the input contributes to that volume.

22.2 4

Perpendicular and equal in length makes it a square: $2 \times 2 = 4$. So their cross product is a vector of length 4.

23 Think faster, talk smarter — structure under pressure

23.1 C — Feeling that you must be interesting

The pressure to be interesting is the problem; being interested is the fix.

23.2 A — Answer, give a Detailed example, Describe why it matters

He uses it for Q&A — answer, example, relevance — and demonstrates it on a mock interview question.

23.3 D — Finishing each thought out of breath, so the inhale becomes a pause

You cannot say 'um' while breathing in, so the pause replaces the filler.

24 Visualizing the chain rule and product rule

24.1 B — Nudging x grows the box along two sides, adding one thin rectangle on each

Each side length is its own function of x, so each changes, and each change contributes a strip. The corner piece carries dx^2 and vanishes, which is why there are exactly two terms and not three.

24.2 6

With $h = x^2$: $\frac{dg}{dh} = 3h^2 = 3x^4$ and $\frac{dh}{dx} = 2x$, so the product is $6x^5$, which is 6 at $x = 1$. Expanding first gives x^6 and the same answer.

25 What's so special about Euler's number e?

25.1 B — 2^{t+dt} factors as $2^t \cdot 2^{dt}$, so everything depending on dt separates from t entirely

After factoring, the dt-dependent piece $\frac{2^{dt}-1}{dt}$ contains no t, so it approaches a fixed number no matter where you started. The derivative is therefore that constant times the original function.

25.2 3

Each constant is the natural log of the base, and $\ln 8 = \ln 2^3 = 3 \ln 2$. So $2.079/0.6931 = 3$ exactly — the pattern is a law of logarithms, not a coincidence.

26 Implicit differentiation, what's going on here?

26.1 B — The step (dx, dy) keeps the expression's value unchanged, so it stays on the curve

$2x\,dx + 2y\,dy$ is how much $x^2 + y^2$ changes under an arbitrary step. Requiring that change to be zero is requiring the point to remain at the same value — which is what being on the curve means.

26.2 −0.75

$2x\,dx + 2y\,dy = 0$ gives $dy/dx = -x/y = -3/4$. The tangent is perpendicular to the radius, which the geometry confirms independently.

27 Story Proofs, Axioms of Probability

27.1 B — Because nothing distinguishes the teams, so each split is counted once as 'this five' and once as 'the other five'

With a team of 4 and a team of 6, naming one names the other unambiguously. Two equal unlabelled teams have no such asymmetry, so every split appears twice.

27.2 126

$\binom{10}{5}/2 = 252/2 = 126$. Equivalently, fix person 1 onto a team and choose their four teammates: $\binom{9}{4} = 126$.

27.3 56

Vandermonde with $m = 3$, $n = 5$, $k = 5$ gives $\binom{8}{5} = 56$. The terms with $j > 3$ vanish, since $\binom{3}{j} = 0$ there.

28 Makemore 1 — bigram character-level language model

28.1 B — A start only ever opens a bigram and an end only closes one, so they never clash

A start token only ever appears as the first character of a bigram and an end token only as the second, so one symbol can play both roles without ambiguity. With two tokens the table had a row for the end token and a column for the start token that could only ever hold zeros.

28.2 9

'.isabella.' has 10 symbols, so 9 adjacent pairs: one for the first letter, seven between letters, and one for the end. In general a name of n letters gives n + 1 examples — 'emma' gives the 5 shown in the notebook.

28.3 D — Without keepdims the sums broadcast as a row, and columns get divided

P.sum(1, keepdims=True) is 27 × 1, which stretches across the columns, so row i is divided by row i's total. P.sum(1) is shape 27; broadcasting aligns it from the right as 1 × 27, so column j is divided by row j's total — silently wrong.

29 The Pixar story spine — six prompts that hold a story up

29.1 C — Because of that

If 'and then' would work just as well, the story has gone slack at that point.

29.2 A — A wireframe or outline of the whole story

Details come afterwards, chosen to push toward the ending.

29.3 D — Going to the grocery store

Anything in your life can be structured as a story spine; it suggests five to ten minutes on something that small.

30 How to easily be funnier in conversations

30.1 A — A true answer that reveals your values, and a playful one

They are free gimmes, like an uncle's ready-made joke.

30.2 B — Humour is often just the unexpected happening

It is not the funniest kind of humour, but it is the easiest and most reliable.

30.3 C — Watching ten minutes of a comedian whose vibe you want, before you go

It uses recency on yourself: the demeanour carries over even when the lines do not.

31 Cramer's rule, explained geometrically

31.1 B — Scales every area by the same factor, its determinant

One common factor for every area is what lets you divide it back out. Preserving areas would make the determinant 1, which is a special case, not the rule.

31.2 2

$\det \begin{pmatrix} 9 & 1 \\ 8 & 2 \end{pmatrix} = 18 - 8 = 10$, divided by $\det A = 6 - 1 = 5$.

32 Change of basis

32.1 B — The same transformation M, described in the other basis

Translate in, transform, translate back. The transformation is unchanged; only the language describing it has moved.

32.2 −4

$-1 \cdot 2 + 2 \cdot (-1) = -2 - 2 = -4$.

33 Eigenvectors and eigenvalues

33.1 B — A nonzero vector can only be sent to zero if the map collapses space

$(A - \lambda I)v = 0$ with $v \neq 0$ requires a nontrivial null space, which means zero determinant.

33.2 3

Triangular, so the eigenvalues are the diagonal entries 3 and 2.

34 A quick trick for computing eigenvalues

34.1 B — The determinant

p is the product of the eigenvalues, and the product of the eigenvalues is the determinant. m is the mean, which comes from the trace.

34.2 9

Mean of the diagonal is 5; determinant is $16 - 7 = 9$. So $\lambda = 5 \pm \sqrt{25 - 9} = 5 \pm 4$, giving 9 and 1.